Mathematics: founded on Set Theory

O Chansol & the Echoes

November 2026

Foundations

Set Theory

Prerequisites of Set Theory

This theory builds on the theory developed in Logic and Language.

1 Sets

1.1 $\forall A((A\mathbf{isASet}) \diamond \nabla)$ *d.a.r.*

1.2 $\forall A(A\mathbf{isASet})$ *t.*

1	∇	T, Logic ??.??
2	$\forall A((A\mathbf{isASet}) \diamond \nabla)$	DA, Set 1.1
3	$((A^{\cdot}\mathbf{isASet}) \diamond \nabla)$	$\forall$E, 2
4	$(((A^{\cdot}\mathbf{isASet}) > \nabla)$ $(\nabla > (A^{\cdot}\mathbf{isASet})))$	D$\diamond$E, 3
5	$(\nabla > (A^{\cdot}\mathbf{isASet}))$	$\wedge$E, 4
6	$(A^{\cdot}\mathbf{isASet})$	$>$E, 5, 1
7	$\forall A(A\mathbf{isASet})$	$\forall$E, 6

1.3 Binary relation symbol **isAnElementOf** is a primitive notion. *d.a.r.*
$\forall A((A\mathbf{isASet}) > \forall B((B\mathbf{isASet}) > ((A \ni B) \diamond (A\mathbf{isAnElementOf}B))))$

1.4 $\forall A((A\mathbf{isASet}) > \forall B((B\mathbf{isASet}) > ((A\mathbf{isAContainerOf}B) \diamond (B\mathbf{isAnElementOf}A)))))$ *d.a.r.*
$\forall A((A\mathbf{isASet}) > \forall B((B\mathbf{isASet}) > ((A \in B) \diamond (A\mathbf{isAContainerOf}B))))$

1.5 $\forall A((A\mathbf{isASet}) > \forall B((B\mathbf{isASet}) > (\forall C((C\mathbf{isASet}) > ((C \ni A) \diamond (C \ni B))) > (A = B))))$ *a.*

1.6 $\forall A\forall B((A = B) \diamond \forall C((C \ni A) \diamond (C \ni B)))$ *t.*

1	$(A^{\cdot} = B^{\cdot})$	
2	$\forall A\forall B((A = B) > ((C^{\cdot} \ni A) \diamond (C^{\cdot} \ni B)))$	AS, Logic ??.??
3	$\forall B((A^{\cdot} = B) > ((C^{\cdot} \ni A^{\cdot}) \diamond (C^{\cdot} \ni B)))$	$\forall$E, 2
4	$((A^{\cdot} = B^{\cdot}) > ((C^{\cdot} \ni A^{\cdot}) \diamond (C^{\cdot} \ni B^{\cdot})))$	$\forall$E, 3

5	$((C^{\cdot} \ni A^{\cdot}) \diamond (C^{\cdot} \ni B^{\cdot}))$	>E, 4, 1
6	$\forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot}))$	∀I, 5
7	$((A^{\cdot} = B^{\cdot}) > \forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot})))$	>I, 1–6
8	$\forall A \forall B(\forall C((C \ni A) \diamond (C \ni B)) > (A = B))$	A, Set 1.5
9	$\forall B(\forall C((C \ni A^{\cdot}) \diamond (C \ni B)) > (A^{\cdot} = B))$	∀E, 8
10	$(\forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot})) > (A^{\cdot} = B^{\cdot}))$	∀E, 9
11	$(((A^{\cdot} = B^{\cdot}) > \forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot})))\wedge$ $(\forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot})) > (A^{\cdot} = B^{\cdot})))$	∧I, 7, 10
12	$((A^{\cdot} = B^{\cdot}) \diamond \forall C((C \ni A^{\cdot}) \diamond (C \ni B^{\cdot})))$	D◇I, 11
13	$\forall B((A^{\cdot} = B) \diamond \forall C((C \ni A^{\cdot}) \diamond (C \ni B)))$	∀I, 12
14	$\forall A \forall B((A = B) \diamond \forall C((C \ni A) \diamond (C \ni B)))$	∀I, 13

1.7 Uniqueness of a set specified by its elements theorem schema

For each formula $\phi[x]$ in which B and C do not occur free. Additional parameters are permitted; bound variables are renamed to avoid capture.

0	$(\forall x((x \ni B^{\cdot}) \diamond \phi[x]) \wedge \forall x((x \ni C^{\cdot}) \diamond \phi[x]))$	
1	$\forall x((x \ni B^{\cdot}) \diamond \phi[x])$	∧E (0)
10	$\forall x((x \ni C^{\cdot}) \diamond \phi[x])$	∧E (0)
11	$((x^{\cdot} \ni B^{\cdot}) \diamond \phi[x^{\cdot}])$	∀E (1)
100	$((x^{\cdot} \ni C^{\cdot}) \diamond \phi[x^{\cdot}])$	∀E (10)
101	$((x^{\cdot} \ni B^{\cdot}) \diamond (x^{\cdot} \ni C^{\cdot}))$	D◇E, >I, >E, D◇I (11, 100)

110	$\forall x((x \ni B^{\cdot}) \diamond (x \ni C^{\cdot}))$	∀I (101)
111	$(B^{\cdot} = C^{\cdot})$	T 1.6, ∀E, D⋄E, >E (110)
1000	$\forall B \forall C((\forall x((x \ni B) \diamond \phi[x]) \wedge \forall x((x \ni C) \diamond \phi[x])) > (B = C))$	>I, ∀I (0–111)
1001	$\exists B \forall x((x \ni B) \diamond \phi[x])$	
1010	$\exists^{\times} B \forall x((x \ni B) \diamond \phi[x])$	∧I, D∃×I (1001, 1000)
1011	$(\exists B \forall x((x \ni B) \diamond \phi[x]) > \exists^{\times} B \forall x((x \ni B) \diamond \phi[x]))$	>I (1001–1010)

$(\exists B \forall x((x \ni B) \diamond \phi[x]) > \exists^{\times} B \forall x((x \ni B) \diamond \phi[x]))$

1.8 A set being a subset of a set defining axiom for a relation symbol

$\forall A((A\cdot) > \forall B((B\cdot) > ((A \mathbf{isASubsetOf} B) \diamond \forall x((x \ni A) > (x \ni B)))))$

$\forall A((A\cdot) > \forall B((B\cdot) > ((A \supset B) \diamond \forall x((x \ni A) > (x \ni B)))))$

1.9 Reflexivity of inclusion theorem

0	$(x^{\cdot} \ni A^{\cdot})$	
1	$(x^{\cdot} \ni A^{\cdot})$	R (0)
10	$\forall x((x \ni A^{\cdot}) > (x \ni A^{\cdot}))$	>I, ∀I (0–1)
11	$(A^{\cdot} \supset A^{\cdot})$	DA 1.8, T 1.2, D⋄E, >E (10)
100	$\forall A((A\cdot) > (A \supset A))$	>I, ∀I (11)

$\forall A((A\cdot) > (A \supset A))$

1.10 Transitivity of inclusion theorem

0	$((A^{\cdot} \supset B^{\cdot}) \wedge (B^{\cdot} \supset C^{\cdot}))$	
1	$\forall x((x \ni A^{\cdot}) > (x \ni B^{\cdot}))$	DA 1.8, ∧E, D⋄E (0)
10	$\forall x((x \ni B^{\cdot}) > (x \ni C^{\cdot}))$	DA 1.8, ∧E, D⋄E (0)
11	$(x^{\cdot} \ni A^{\cdot})$	
100	$(x^{\cdot} \ni B^{\cdot})$	∀E, >E (1, 11)
101	$(x^{\cdot} \ni C^{\cdot})$	∀E, >E (10, 100)

110	$\forall x((x \ni A^{\cdot}) > (x \ni C^{\cdot}))$	>I, ∀I (11–101)
111	$(A^{\cdot} \supset C^{\cdot})$	DA 1.8, D⋄E (110)
1000	$\forall A \forall B \forall C(((A \supset B) \wedge (B \supset C)) > (A \supset C))$	>I, ∀I (0–111)

$\forall A \forall B \forall C(((A \supset B) \wedge (B \supset C)) > (A \supset C))$

1.11 Antisymmetry of inclusion theorem

0	$((A^{\cdot} \supset B^{\cdot}) \wedge (B^{\cdot} \supset A^{\cdot}))$	
1	$\forall x((x \ni A^{\cdot}) \diamond (x \ni B^{\cdot}))$	DA 1.8, D⋄I (0)
10	$(A^{\cdot} = B^{\cdot})$	T 1.6, D⋄E (1)
11	$\forall A \forall B(((A \supset B) \wedge (B \supset A)) > (A = B))$	>I, ∀I (0–10)

$\forall A \forall B(((A \supset B) \wedge (B \supset A)) > (A = B))$

1.12 The axiom of empty set axiom

$\exists A((A\cdot) \wedge \forall x((x\cdot) > \backslash(x \ni A)))$

1.13 The empty set defining axiom for a function symbol

0	$\exists A \forall x \backslash (x \ni A)$	A 1.12, T 1.2
1	$\forall A (\forall x \backslash (x \ni A) \diamond \forall x ((x \ni A) \diamond \triangle))$	D\E, D$\diamond$I
10	$\exists^{\times} A \forall x \backslash (x \ni A)$	TS 1.7, D$\exists^{\times}$I (0, 1)

$\forall x ((x \cdot) > \backslash (x \ni (\textbf{theEmptySet})))$

$\forall x ((x \cdot) > \backslash (x \ni \bigcirc))$

1.14 The axiom schema of specification

axiom schema

For each formula $\phi[x]$ with B not free. Additional parameters are permitted; bound variables are renamed to avoid capture.

$\forall A ((A \cdot) > \exists B ((B \cdot) \wedge \forall x ((x \cdot) > ((x \ni B) \diamond ((x \ni A) \wedge \phi[x])))))$

1.15 The specified subset of a set

defining axiom schema for a function symbol

For each formula $\phi[x]$ with B not free, under the same variable conditions as specification.

0	$\forall A \exists B \forall x ((x \ni B) \diamond ((x \ni A) \wedge \phi[x]))$	AS 1.14, T 1.2
1	$\forall A \exists^{\times} B \forall x ((x \ni B) \diamond ((x \ni A) \wedge \phi[x]))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A ((A \cdot) > \forall x ((x \cdot) > ((x \ni (\textbf{theSetSpecifiedBy}\phi\textbf{From}A)) \diamond ((x \ni A) \wedge \phi[x]))))$

$\forall A ((A \cdot) > \forall x ((x \cdot) > ((x \ni \{y \ni A \mid \phi[y]\}) \diamond ((x \ni A) \wedge \phi[x]))))$

1.16 The axiom of binding

axiom

$\forall A ((A \cdot) > \forall B ((B \cdot) > \exists C ((C \cdot) \wedge \forall x ((x \cdot) > ((x \ni C) \diamond ((x = A) \vee (x = B)))))))$

1.17 The binding of two sets

defining axiom for a function symbol

0	$\forall A \forall B \exists C \forall x ((x \ni C) \diamond ((x = A) \vee (x = B)))$	A 1.16, T 1.2
1	$\forall A \forall B \exists^{\times} C \forall x ((x \ni C) \diamond ((x = A) \vee (x = B)))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x\cdot) > ((x \ni (\mathbf{theBindingOf}\, A\, \mathbf{And}\, B)) \diamond ((x = A) \vee (x = B))))))$

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x\cdot) > ((x \ni \{A, B\}) \diamond ((x = A) \vee (x = B))))))$

1.18 The singleton of a set

defining axiom for a function symbol

0	$\forall A \forall x((x \ni \{A, A\}) \diamond ((x = A) \vee (x = A)))$	DA 1.17, T 1.2
1	$\forall A \forall x((x \ni \{A, A\}) \diamond (x = A))$	$\vee$I, $\vee$E, D$\diamond$I (0)
10	$\forall A \exists^{\times} B \forall x((x \ni B) \diamond (x = A))$	TS 1.7, $\exists$I (1)

$\forall A((A\cdot) > \forall x((x\cdot) > ((x \ni (\mathbf{theSingletonOf}\, A)) \diamond (x = A))))$

$\forall A((A\cdot) > \forall x((x\cdot) > ((x \ni \{A\}) \diamond (x = A))))$

1.19 The axiom of union

axiom

$\forall A((A\cdot) > \exists B((B\cdot) \wedge \forall x((x\cdot) > ((x \ni B) \diamond \exists C((C\cdot) \wedge ((C \ni A) \wedge (x \ni C)))))))$

1.20 The union of sets in a set

defining axiom for a function symbol

0	$\forall A \exists B \forall x((x \ni B) \diamond \exists C((C \ni A) \wedge (x \ni C)))$	A 1.19, T 1.2
1	$\forall A \exists^{\times} B \forall x((x \ni B) \diamond \exists C((C \ni A) \wedge (x \ni C)))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A((A\cdot) > \forall x((x\cdot) > ((x \ni (\mathbf{theUnionOfSetsIn}\, A)) \diamond \exists B((B \ni A) \wedge (x \ni B)))))$

$\forall A((A\cdot) > \forall x((x\cdot) > ((x \ni \bigcup A) \diamond \exists B((B \ni A) \wedge (x \ni B)))))$

1.21 The union of two sets

defining axiom for a function symbol

0	$\forall x((x \ni \bigcup\{A^{\cdot}, B^{\cdot}\}) \diamond \exists C((C \ni \{A^{\cdot}, B^{\cdot}\}) \wedge (x \ni C)))$	DA 1.20
1	$\forall x((x \ni \bigcup\{A^{\cdot}, B^{\cdot}\}) \diamond ((x \ni A^{\cdot}) \vee (x \ni B^{\cdot})))$	DA 1.17, $\exists$E, $\vee$E, $\exists$I (0)
10	$\forall A \forall B \exists^{\times} C \forall x((x \ni C) \diamond ((x \ni A) \vee (x \ni B)))$	TS 1.7, $\exists$I, $\forall$I (1)

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (\mathbf{theUnionOf}\, A\, \mathbf{And}\, B)) \diamond ((x \ni A) \vee (x \ni B)))))$

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (A \cup B)) \diamond ((x \ni A) \vee (x \ni B)))))$

1.22 The intersection of two sets defining axiom for a function symbol

0	$\forall A \forall B \exists C \forall x((x \ni C) \diamond ((x \ni A) \wedge (x \ni B)))$	AS 1.14, T 1.2
1	$\forall A \forall B \exists^{\times} C \forall x((x \ni C) \diamond ((x \ni A) \wedge (x \ni B)))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (\mathbf{theIntersectionOf}\, A\, \mathbf{And}\, B)) \diamond ((x \ni A) \wedge (x \ni B)))))$

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (A \cap B)) \diamond ((x \ni A) \wedge (x \ni B)))))$

1.23 The difference of two sets defining axiom for a function symbol

0	$\forall A \forall B \exists C \forall x((x \ni C) \diamond ((x \ni A) \wedge \backslash(x \ni B)))$	AS 1.14, T 1.2
1	$\forall A \forall B \exists^{\times} C \forall x((x \ni C) \diamond ((x \ni A) \wedge \backslash(x \ni B)))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (\mathbf{theDifferenceOf}\, A\, \mathbf{And}\, B)) \diamond ((x \ni A) \wedge \backslash(x \ni B)))))$

$\forall A((A\cdot) > \forall B((B\cdot) > \forall x((x \ni (A \sim B)) \diamond ((x \ni A) \wedge \backslash(x \ni B)))))$

1.24 The axiom of power set axiom

$\forall A((A\cdot) > \exists B((B\cdot) \wedge \forall C((C\cdot) > ((C \ni B) \diamond (C \supset A)))))$

1.25 The power set of a set defining axiom for a function symbol

0	$\forall A \exists B \forall C((C \ni B) \diamond (C \supset A))$	A 1.24, T 1.2
1	$\forall A \exists^{\times} B \forall C((C \ni B) \diamond (C \supset A))$	TS 1.7, $\exists^{\times}$I (0)

$\forall A((A\cdot) > \forall B((B\cdot) > ((B \ni (\mathbf{thePowerSetOf}\, A)) \diamond (B \supset A))))$

$\forall A((A\cdot) > \forall B((B\cdot) > ((B \ni \mathcal{P}(A)) \diamond (B \supset A))))$

1.26 The axiom schema of substitution axiom schema

For each formula $\phi[x, y]$ with B not free. Additional parameters are permitted; all substitutions avoid variable capture.

$\forall A((A\cdot) > (\forall x \forall y \forall z(((\phi[x, y] \wedge \phi[x, z]) > (y = z))) >$

$\exists B((B\cdot) \wedge \forall y((y \ni B) \diamond \exists x((x \ni A) \wedge \phi[x, y])))))$

1.27 The axiom of regularity axiom

$\forall A((A\cdot) > (\backslash(A = \bigcirc) > \exists B((B \ni A) \wedge \backslash\exists x((x \ni A) \wedge (x \ni B)))))$

1.28 A set being nonempty defining axiom for a relation symbol

$\forall A((A\cdot) > ((A\mathbf{isNonempty}) \diamond \exists x(x \ni A)))$

1.29 Emptiness and membership theorem

0	$\forall A((A = \bigcirc) \diamond \forall x((x \ni A) \diamond (x \ni \bigcirc)))$	T 1.6
1	$\forall A((A = \bigcirc) \diamond \forall x \backslash(x \ni A))$	DA 1.13, D$\diamond$E, D\E (0)
10	$\forall A((A = \bigcirc) \diamond \backslash\exists x(x \ni A))$	$\forall$E, $\exists$E, D\I (1)

$\forall A((A = \bigcirc) \diamond \backslash\exists x(x \ni A))$

2 Pairs

2.1 The pair of two sets defining axiom for a function symbol

0	$(\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\}=\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\})$	=I
1	$\exists X(X=\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\})$	∃I (0)
10	$\forall X\forall Y(((X=\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\})\wedge(Y=\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\}))>(X=Y))$	AS, =I
11	$\exists^{\times}X(X=\{\{A^{\cdot}\},\{A^{\cdot},B^{\cdot}\}\})$	D∃×I (1, 10)

$\forall A((A\cdot)>\forall B((B\cdot)>((\mathbf{thePairOf}\,A\,\mathbf{And}\,B)=\{\{A\},\{A,B\}\})))$

$\forall A((A\cdot)>\forall B((B\cdot)>(\langle A,B\rangle=\{\{A\},\{A,B\}\})))$

2.2 The common elements of the members of a pair theorem

0	$\forall Z((Z\ni\langle A^{\cdot},B^{\cdot}\rangle)>(x^{\cdot}\ni Z))$	
1	$(\{A^{\cdot}\}\ni\langle A^{\cdot},B^{\cdot}\rangle)$	DA 2.1, DA 1.17
10	$(x^{\cdot}\ni\{A^{\cdot}\})$	∀E, >E (0, 1)
11	$(x^{\cdot}=A^{\cdot})$	DA 1.18 (10)
100	$(\forall Z((Z\ni\langle A^{\cdot},B^{\cdot}\rangle)>(x^{\cdot}\ni Z))>(x^{\cdot}=A^{\cdot}))$	>I (0–11)
101	$(x^{\cdot}=A^{\cdot})$	

110	$((x^{\cdot}\ni\{A^{\cdot}\})\wedge(x^{\cdot}\ni\{A^{\cdot},B^{\cdot}\}))$	DA 1.18, DA 1.17, AS (101)
111	$\forall Z((Z\ni\langle A^{\cdot},B^{\cdot}\rangle)>(x^{\cdot}\ni Z))$	DA 2.1, ∨E, >I, ∀I (110)
1000	$\forall A\forall B\forall x(\forall Z((Z\ni\langle A,B\rangle)>(x\ni Z))\diamond(x=A))$	>I, D⋄I, ∀I (100, 101–111)

$\forall A\forall B\forall x(\forall Z((Z\ni\langle A,B\rangle)>(x\ni Z))\diamond(x=A))$

2.3 The union of a pair theorem

0	$\forall x((x \ni \bigcup\langle A^{\cdot}, B^{\cdot}\rangle) \diamond ((x = A^{\cdot}) \vee (x = B^{\cdot})))$	DA 1.20, DA 2.1, DA 1.18, DA 1.17
1	$\forall x((x \ni \bigcup\langle A^{\cdot}, B^{\cdot}\rangle) \diamond (x \ni \{A^{\cdot}, B^{\cdot}\}))$	DA 1.17, D$\diamond$I (0)
10	$\forall A \forall B(\bigcup\langle A, B\rangle = \{A, B\})$	T 1.6, $\forall$I (1)

$\forall A \forall B(\bigcup\langle A, B\rangle = \{A, B\})$

2.4 Cancellation of a common member in equal bindings theorem

0	$(\{A^{\cdot}, B^{\cdot}\} = \{A^{\cdot}, C^{\cdot}\})$	
1	$((B^{\cdot} = A^{\cdot}) \vee (B^{\cdot} = C^{\cdot}))$	DA 1.17, AS (0)
10	$((C^{\cdot} = A^{\cdot}) \vee (C^{\cdot} = B^{\cdot}))$	DA 1.17, AS (0)
11	$(B^{\cdot} = A^{\cdot})$	
100	$(B^{\cdot} = C^{\cdot})$	AS, $\vee$E (10, 11)
101	$((B^{\cdot} = A^{\cdot}) > (B^{\cdot} = C^{\cdot}))$	$>$I (11–100)

110	$(B^{\cdot} = C^{\cdot})$	
111	$(B^{\cdot} = C^{\cdot})$	R (110)
1000	$((B^{\cdot} = C^{\cdot}) > (B^{\cdot} = C^{\cdot}))$	$>$I (110–111)
1001	$(B^{\cdot} = C^{\cdot})$	$\vee$E (1, 101, 1000)
1010	$\forall A \forall B \forall C((\{A, B\} = \{A, C\}) > (B = C))$	$>$I, $\forall$I (0–1001)

$\forall A \forall B \forall C((\{A, B\} = \{A, C\}) > (B = C))$

2.5 Equality of pairs theorem

0	$(\langle A^{\cdot}, B^{\cdot}\rangle = \langle C^{\cdot}, D^{\cdot}\rangle)$	
1	$\forall x((x = A^{\cdot}) \diamond (x = C^{\cdot}))$	T 2.2, AS (0)
10	$(A^{\cdot} = C^{\cdot})$	∀E, =I, D⋄E (1)
11	$(\{A^{\cdot}, B^{\cdot}\} = \{C^{\cdot}, D^{\cdot}\})$	T 2.3, AS (0)
100	$(\{A^{\cdot}, B^{\cdot}\} = \{A^{\cdot}, D^{\cdot}\})$	AS (10, 11)
101	$(B^{\cdot} = D^{\cdot})$	T 2.4, ∀E, >E (100)

110	$((A^{\cdot} = C^{\cdot}) \wedge (B^{\cdot} = D^{\cdot}))$	∧I (10, 101)
111	$((\langle A^{\cdot}, B^{\cdot}\rangle = \langle C^{\cdot}, D^{\cdot}\rangle) > ((A^{\cdot} = C^{\cdot}) \wedge (B^{\cdot} = D^{\cdot})))$	>I (0–110)
1000	$(((A^{\cdot} = C^{\cdot}) \wedge (B^{\cdot} = D^{\cdot})) > (\langle A^{\cdot}, B^{\cdot}\rangle = \langle C^{\cdot}, D^{\cdot}\rangle))$	AS, =I, >I
1001	$\forall A \forall B \forall C \forall D((\langle A, B\rangle = \langle C, D\rangle) \diamond ((A = C) \wedge (B = D)))$	D⋄I, ∀I (111, 1000)

$\forall A \forall B \forall C \forall D((\langle A, B\rangle = \langle C, D\rangle) \diamond ((A = C) \wedge (B = D)))$

2.6 A set being a pair defining axiom for a relation symbol

$\forall P((P\cdot) > ((P\mathbf{isAPair}) \diamond \exists A((A\cdot) \wedge \exists B((B\cdot) \wedge (P = \langle A, B\rangle)))))$

2.7 The first projection of a pair defining axiom for a function symbol

0	$\forall P((P\mathbf{isAPair}) > \exists X(\exists B(P = \langle X, B\rangle)))$	DA 2.6, ∃E, ∃I
1	$\forall P \forall X \forall Y(((\exists B(P = \langle X, B\rangle)) \wedge (\exists B(P = \langle Y, B\rangle))) > (X = Y))$	T 2.5, ∃E
10	$\forall P((P\mathbf{isAPair}) > \exists^{\times} X(\exists B(P = \langle X, B\rangle)))$	D∃×I (0, 1)

$\forall P((P\cdot) > ((P\mathbf{isAPair}) > \exists B(P = \langle(\mathbf{theFirstProjectionOf}P), B\rangle)))$

$\forall P((P\cdot) > ((P\mathbf{isAPair}) > \exists B(P = \langle(\pi_0 P), B\rangle)))$

2.8 The second projection of a pair defining axiom for a function symbol

0	$\forall P((P\mathbf{isAPair}) > \exists X(\exists A(P = \langle A, X\rangle)))$	DA 2.6, ∃E, ∃I
1	$\forall P\forall X\forall Y(((\exists A(P = \langle A, X\rangle)) \wedge (\exists A(P = \langle A, Y\rangle))) > (X = Y))$	T 2.5, ∃E
10	$\forall P((P\mathbf{isAPair}) > \exists^{\times} X(\exists A(P = \langle A, X\rangle)))$	D∃×I (0, 1)

$\forall P((P\cdot) > ((P\mathbf{isAPair}) > \exists A(P = \langle A, (\mathbf{theSecondProjectionOf}\, P)\rangle)))$

$\forall P((P\cdot) > ((P\mathbf{isAPair}) > \exists A(P = \langle A, (\pi_1 P)\rangle)))$

2.9 The projections of a pair theorem

0	$(\langle A^{\cdot}, B^{\cdot}\rangle \mathbf{isAPair})$	DA 2.6, =I, ∃I
1	$\exists Y(\langle A^{\cdot}, B^{\cdot}\rangle = \langle(\pi_0\langle A^{\cdot}, B^{\cdot}\rangle), Y\rangle)$	DA 2.7 (0)
10	$((\pi_0\langle A^{\cdot}, B^{\cdot}\rangle) = A^{\cdot})$	T 2.5, ∃E (1)
11	$\exists X(\langle A^{\cdot}, B^{\cdot}\rangle = \langle X, (\pi_1\langle A^{\cdot}, B^{\cdot}\rangle)\rangle)$	DA 2.8 (0)
100	$((\pi_1\langle A^{\cdot}, B^{\cdot}\rangle) = B^{\cdot})$	T 2.5, ∃E (11)
101	$\forall A\forall B(((\pi_0\langle A, B\rangle) = A) \wedge ((\pi_1\langle A, B\rangle) = B))$	∧I, ∀I (10, 100)

$\forall A\forall B(((\pi_0\langle A, B\rangle) = A) \wedge ((\pi_1\langle A, B\rangle) = B))$

2.10 Bounds for the components of a pair theorem

0	$\forall A\forall B(\bigcup\langle A, B\rangle = \{A, B\})$	T 2.3
1	$\forall A\forall B((A \ni \bigcup\langle A, B\rangle) \wedge (B \ni \bigcup\langle A, B\rangle))$	DA 1.17 (0)
10	$((A^{\cdot} \ni S^{\cdot}) \wedge (B^{\cdot} \ni S^{\cdot}))$	
11	$((\{A^{\cdot}\} \supset S^{\cdot}) \wedge (\{A^{\cdot}, B^{\cdot}\} \supset S^{\cdot}))$	DA 1.18, DA 1.17, DA 1.8 (10)
100	$(\langle A^{\cdot}, B^{\cdot}\rangle \supset \mathcal{P}(S^{\cdot}))$	DA 2.1, DA 1.25 (11)
101	$(\langle A^{\cdot}, B^{\cdot}\rangle \ni \mathcal{P}(\mathcal{P}(S^{\cdot})))$	DA 1.25 (100)

110	$\forall A\forall B\forall S(((A \ni S) \wedge (B \ni S)) > (\langle A, B\rangle \ni \mathcal{P}(\mathcal{P}(S))))$	>I, ∀I (10–101)

$\forall A \forall B((A \ni \bigcup \langle A, B \rangle) \wedge (B \ni \bigcup \langle A, B \rangle))$

$\forall A \forall B \forall S(((A \ni S) \wedge (B \ni S)) > (\langle A, B \rangle \ni \mathcal{P}(\mathcal{P}(S))))$

3 Relations

3.1 A set being a relation defining axiom for a relation symbol

$\forall R((R\cdot) > ((R\mathbf{isARelation}) \diamond \forall P((P\cdot) > ((P \ni R) > (P\mathbf{isAPair})))))$

$\forall R((R\cdot) > ((R\mathbf{isASetOfPairs}) \diamond \forall P((P\cdot) > ((P \ni R) > (P\mathbf{isAPair})))))$

3.2 The domain of a relation defining axiom for a function symbol

0	$(R^{\cdot}\mathbf{isARelation})$	
1	$\forall x(\exists P((P \ni R^{\cdot}) \wedge ((\pi_0 P) = x)) > (x \ni \bigcup\bigcup R^{\cdot}))$	DA 3.1, T 2.10, T 2.9, DA 1.20 (0)
10	$\exists D \forall x((x \ni D) \diamond \exists P((P \ni R^{\cdot}) \wedge ((\pi_0 P) = x)))$	AS 1.14 (1)
11	$\exists^{\times} D \forall x((x \ni D) \diamond \exists P((P \ni R^{\cdot}) \wedge ((\pi_0 P) = x)))$	TS 1.7 (10)
100	$\forall R((R\mathbf{isARelation}) > \exists^{\times} D \forall x((x \ni D) \diamond \exists P((P \ni R) \wedge ((\pi_0 P) = x))))$	>I, ∀I (0–11)

$\forall R((R\cdot) > ((R\mathbf{isARelation}) > \forall x((x \ni (\mathbf{theDomainOf}\, R)) \diamond \exists P((P \ni R) \wedge ((\pi_0 P) = x)))))$

3.3 The range of a relation defining axiom for a function symbol

0	$(R^{\cdot}\mathbf{isARelation})$	
1	$\forall y(\exists P((P \ni R^{\cdot}) \wedge ((\pi_1 P) = y)) > (y \ni \bigcup\bigcup R^{\cdot}))$	DA 3.1, T 2.10, T 2.9, DA 1.20 (0)
10	$\exists D \forall y((y \ni D) \diamond \exists P((P \ni R^{\cdot}) \wedge ((\pi_1 P) = y)))$	AS 1.14 (1)
11	$\exists^{\times} D \forall y((y \ni D) \diamond \exists P((P \ni R^{\cdot}) \wedge ((\pi_1 P) = y)))$	TS 1.7 (10)
100	$\forall R((R\mathbf{isARelation}) > \exists^{\times} D \forall y((y \ni D) \diamond \exists P((P \ni R) \wedge ((\pi_1 P) = y))))$	>I, ∀I (0–11)

$\forall R((R\cdot) > ((R\mathbf{isARelation}) > \forall y((y \ni (\mathbf{theRangeOf}\, R)) \diamond \exists P((P \ni R) \wedge ((\pi_1 P) = y)))))$

3.4 The field of a relation defining axiom for a function symbol

0	$(((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot})) = ((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot})))$	=I
1	$\exists X(X = ((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot})))$	$\exists$I (0)
10	$\forall X \forall Y(((X = ((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot}))) \wedge (Y = ((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot})))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = ((\mathbf{theDomainOf}R^{\cdot}) \cup (\mathbf{theRangeOf}R^{\cdot})))$	D$\exists^{\times}$I (1, 10)

$\forall R((R\cdot) > ((R\mathbf{isARelation}) > ((\mathbf{theFieldOf}R) = ((\mathbf{theDomainOf}R) \cup (\mathbf{theRangeOf}R)))))$

3.5 A relation being on a set defining axiom for a relation symbol

$\forall R \forall A((R\mathbf{isARelationOn}A)\diamond$

$((R\mathbf{isARelation}) \wedge \forall P((P \ni R) > (((\pi_0 P) \ni A) \wedge ((\pi_1 P) \ni A)))))$

3.6 A relation being reflexive on a set defining axiom for a relation symbol

$\forall R \forall A((R\mathbf{isARelationOn}A) > ((R\mathbf{isReflexiveOn}A) \diamond \forall x((x \ni A) > (\langle x, x\rangle \ni R))))$

3.7 A relation being symmetric defining axiom for a relation symbol

$\forall R((R\mathbf{isARelation}) > ((R\mathbf{isSymmetric}) \diamond \forall x \forall y((\langle x, y\rangle \ni R) > (\langle y, x\rangle \ni R))))$

3.8 A relation being transitive defining axiom for a relation symbol

$\forall R((R\mathbf{isARelation}) > ((R\mathbf{isTransitive}) \diamond \forall x \forall y \forall z((((\langle x, y\rangle \ni R) \wedge (\langle y, z\rangle \ni R)) > (\langle x, z\rangle \ni R)))))$

3.9 A relation being an equivalence relation on a set defining axiom for a relation symbol

$\forall R \forall A((R\mathbf{isAnEquivalenceRelationOn}A)\diamond$

$((R\mathbf{isARelationOn}A) \wedge ((R\mathbf{isReflexiveOn}A) \wedge ((R\mathbf{isSymmetric}) \wedge (R\mathbf{isTransitive})))))$

3.10 The equivalence class of an element defining axiom for a function symbol

0	$(\{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\} = \{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\})$	=I
1	$\exists X(X = \{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\})$	$\exists$I (0)
10	$\forall X \forall Y(((X = \{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\}) \wedge (Y = \{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{x \ni A^{\cdot} \mid (\langle a^{\cdot}, x\rangle \ni R^{\cdot})\})$	D$\exists^{\times}$I (1, 10)

$\forall R \forall A \forall a(((R\textbf{isAnEquivalenceRelationOn}A) \wedge (a \ni A)) >$

$((\textbf{theEquivalenceClassOf}a\textbf{UnderIn}R A) = \{x \ni A \mid (\langle a, x\rangle \ni R)\}))$

3.11 A relation being antisymmetric defining axiom for a relation symbol

$\forall R((R\textbf{isARelation}) > ((R\textbf{isAntisymmetric}) \diamond \forall x \forall y((((\langle x, y\rangle \ni R) \wedge (\langle y, x\rangle \ni R)) > (x = y))))$

3.12 A relation being a partial order on a set defining axiom for a relation symbol

$\forall R \forall A((R\textbf{isAPartialOrderOn}A) \diamond ((R\textbf{isARelationOn}A) \wedge ((R\textbf{isReflexiveOn}A) \wedge ((R\textbf{isAntisymmetric}) \wedge (R\textbf{isTransitive})))))$

3.13 A relation being a strict partial order on a set defining axiom for a relation symbol

$\forall R \forall A((R\textbf{isAStrictPartialOrderOn}A) \diamond$

$((R\textbf{isARelationOn}A) \wedge ((R\textbf{isTransitive}) \wedge \forall x \forall y((\langle x, y\rangle \ni R) > \backslash(\langle y, x\rangle \ni R)))))$

3.14 A relation being a linear order on a set defining axiom for a relation symbol

$\forall R \forall A((R\textbf{isALinearOrderOn}A) \diamond$

$((R\textbf{isAPartialOrderOn}A) \wedge \forall x((x \ni A) > \forall y((y \ni A) > ((\langle x, y\rangle \ni R) \vee (\langle y, x\rangle \ni R))))))$

3.15 A set being a least element defining axiom for a relation symbol

$\forall R \forall B \forall b((b\textbf{isALeastElementOf}B\textbf{Under}R) \diamond$

$((b \ni B) \wedge \forall x((x \ni B) > (\langle b, x\rangle \ni R))))$

3.16 Uniqueness of a least element theorem

0	$((R^{\cdot}\,\mathbf{isAPartialOrderOn}\,A^{\cdot}) \wedge ((b^{\cdot}\,\mathbf{isALeastElementOf}\,B^{\cdot}\,\mathbf{Under}\,R^{\cdot}) \wedge (c^{\cdot}\,\mathbf{isALeastElementOf}\,B^{\cdot}\,\mathbf{Under}\,R^{\cdot})))$	
1	$((b^{\cdot} \ni B^{\cdot}) \wedge (c^{\cdot} \ni B^{\cdot}))$	DA 3.15, $\wedge$E (0)
10	$((\langle b^{\cdot}, c^{\cdot}\rangle \ni R^{\cdot}) \wedge (\langle c^{\cdot}, b^{\cdot}\rangle \ni R^{\cdot}))$	DA 3.15, $\forall$E, $>$E, $\wedge$I (0, 1)
11	$(b^{\cdot} = c^{\cdot})$	DA 3.12, DA 3.11 (0, 10)
100	$\forall R \forall A \forall B((R\mathbf{isAPartialOrderOn}\,A) > \forall b \forall c(((b\mathbf{isALeastElementOf}\,B\mathbf{Under}\,R) \wedge (c\mathbf{isALeastElementOf}\,B\mathbf{Under}\,R)) > (b = c)))$	$>$I, $\forall$I (0–11)

$\forall R \forall A \forall B((R\mathbf{isAPartialOrderOn}\,A) > \forall b \forall c($

$((b\mathbf{isALeastElementOf}\,B\mathbf{Under}\,R) \wedge (c\mathbf{isALeastElementOf}\,B\mathbf{Under}\,R)) > (b = c)))$

3.17 A relation being a well order on a set defining axiom for a relation symbol

$\forall R \forall A((R\mathbf{isAWellOrderOn}\,A)\diamond$

$((R\mathbf{isALinearOrderOn}\,A) \wedge \forall B(((B \supset A) \wedge \backslash(B = \bigcirc)) >$

$\exists b(b\mathbf{isALeastElementOf}\,B\mathbf{Under}\,R))))$

3.18 The inclusion relation on a set defining axiom for a function symbol

0	$(\{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\} = \{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\})$	=I
1	$\exists X(X = \{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\})$	$\exists$I (0)
10	$\forall X \forall Y(((X = \{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\}) \wedge (Y = \{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{P \ni (A^{\cdot} \times A^{\cdot}) \mid ((\pi_0 P) \supset (\pi_1 P))\})$	D$\exists^{\times}$I (1, 10)

$\forall A((\mathbf{theInclusionRelationOn}\,A) = \{P \ni (A \times A) \mid ((\pi_0 P) \supset (\pi_1 P))\})$

3.19 Inclusion is a partial order theorem

0	$\forall x \forall y((\langle x, y\rangle \ni (\mathbf{theInclusionRelationOn}A^{\cdot})) \diamond (((x \ni A^{\cdot}) \wedge (y \ni A^{\cdot})) \wedge (x \supset y)))$	DA 3.18, DA 1.15, T 7.2, T 2.9
1	$((\mathbf{theInclusionRelationOn}A^{\cdot})\mathbf{isARelationOn}A^{\cdot})$	DA 3.5, DA 3.1, DA 7.1 (0)
10	$((\mathbf{theInclusionRelationOn}A^{\cdot})\mathbf{isReflexiveOn}A^{\cdot})$	T 1.9, DA 3.6 (0, 1)
11	$((\mathbf{theInclusionRelationOn}A^{\cdot})\mathbf{isAntisymmetric})$	T 1.11, DA 3.11 (0, 1)
100	$((\mathbf{theInclusionRelationOn}A^{\cdot})\mathbf{isTransitive})$	T 1.10, DA 3.8 (0, 1)
101	$\forall A((\mathbf{theInclusionRelationOn}A)\mathbf{isAPartialOrderOn}A)$	DA 3.12, ∧I, ∀I (1, 10, 11, 100)

$\forall A((\mathbf{theInclusionRelationOn}A)\mathbf{isAPartialOrderOn}A)$

4 Functions

4.1 A set being a function defining axiom for a relation symbol

$\forall F((F\cdot) > ((F\mathbf{isAFunction}) \diamond ((F\mathbf{isARelation})\wedge$

$\forall P((P \ni F) > \forall Q((Q \ni F) > (((\pi_0 P) = (\pi_0 Q)) > ((\pi_1 P) = (\pi_1 Q))))))))$

4.2 A function is a relation theorem

0	$(F^{\cdot}\mathbf{isAFunction})$	
1	$((F^{\cdot}\mathbf{isARelation}) \wedge \forall P((P \ni F^{\cdot}) > \forall Q((Q \ni F^{\cdot}) > (((\pi_0 P) = (\pi_0 Q)) > ((\pi_1 P) = (\pi_1 Q))))))$	DA 4.1, D$\diamond$E (0)
10	$(F^{\cdot}\mathbf{isARelation})$	$\wedge$E (1)
11	$\forall F((F\mathbf{isAFunction}) > (F\mathbf{isARelation}))$	>I, $\forall$I (0–10)

$\forall F((F\mathbf{isAFunction}) > (F\mathbf{isARelation}))$

4.3 The value of a function at a set defining axiom for a function symbol

0	$((F^{\cdot}\mathbf{isAFunction}) \wedge (x^{\cdot} \ni (\mathbf{theDomainOf}F^{\cdot})))$	
1	$\exists P((P \ni F^{\cdot}) \wedge ((\pi_0 P) = x^{\cdot}))$	T 4.2, DA 3.2 (0)
10	$\exists y(\langle x^{\cdot}, y\rangle \ni F^{\cdot})$	DA 3.1, DA 2.6, T 2.9, $\exists$E, $\exists$I (1)
11	$\forall y\forall z(((\langle x^{\cdot}, y\rangle \ni F^{\cdot}) \wedge (\langle x^{\cdot}, z\rangle \ni F^{\cdot})) > (y = z))$	DA 4.1, T 2.9 (0)
100	$\exists^{\times} y(\langle x^{\cdot}, y\rangle \ni F^{\cdot})$	D$\exists^{\times}$I (10, 11)
101	$\forall F\forall x(((F\mathbf{isAFunction}) \wedge (x \ni (\mathbf{theDomainOf}F))) > \exists^{\times} y(\langle x, y\rangle \ni F))$	>I, $\forall$I (0–100)

$\forall F((F\mathbf{isAFunction}) > \forall x((x \ni (\mathbf{theDomainOf}F)) > (\langle x, (\mathbf{theValueOf}F\mathbf{At}x)\rangle \ni F)))$

$\forall F((F\mathbf{isAFunction}) > \forall x((x \ni (\mathbf{theDomainOf}F)) > (\langle x, (Fx)\rangle \ni F)))$

4.4 A function being from a set to a set defining axiom for a relation symbol

$\forall F \forall A \forall B((F \textbf{ isAFunctionFrom } A \textbf{ To } B) \diamond ((F \textbf{ isAFunction}) \wedge (((\textbf{theDomainOf } F) = A) \wedge ((\textbf{theRangeOf } F) \supset B))))$

4.5 A function being injective defining axiom for a relation symbol

$\forall F((F \textbf{ isAFunction}) > ((F \textbf{ isInjective}) \diamond \forall x((x \ni (\textbf{theDomainOf } F)) >$

$\forall y((y \ni (\textbf{theDomainOf } F)) > (((\textbf{theValueOf } F \textbf{ At } x) = (\textbf{theValueOf } F \textbf{ At } y)) > (x = y))))))$

4.6 A function being surjective onto a set defining axiom for a relation symbol

$\forall F \forall B((F \textbf{ isAFunction}) > ((F \textbf{ isSurjectiveOnto } B) \diamond ((\textbf{theRangeOf } F) = B)))$

4.7 A function being a bijection from a set to a set defining axiom for a relation symbol

$\forall F \forall A \forall B((F \textbf{ isABijectionFrom } A \textbf{ To } B) \diamond$

$((F \textbf{ isAFunctionFrom } A \textbf{ To } B) \wedge ((F \textbf{ isInjective}) \wedge (F \textbf{ isSurjectiveOnto } B))))$

4.8 The identity function on a set defining axiom for a function symbol

0	$\forall x((x \ni A^{\cdot}) > (\langle x, x \rangle \ni \mathcal{P}(\mathcal{P}(A^{\cdot}))))$	T 2.10
1	$\exists F \forall P((P \ni F) \diamond \exists x((x \ni A^{\cdot}) \wedge (P = \langle x, x \rangle)))$	AS 1.14 (0)
10	$\forall A \exists^{\times} F \forall P((P \ni F) \diamond \exists x((x \ni A) \wedge (P = \langle x, x \rangle)))$	TS 1.7, ∀I (1)

$\forall A \forall P((P \ni (\textbf{theIdentityFunctionOn } A)) \diamond \exists x((x \ni A) \wedge (P = \langle x, x \rangle)))$

4.9 Properties of the identity function theorem

0	$\forall P((P \ni (\mathbf{theIdentityFunctionOn}A^{\cdot})) > (P\mathbf{isAPair}))$	DA 4.8, DA 2.6
1	$((\mathbf{theIdentityFunctionOn}A^{\cdot})\mathbf{isARelation})$	DA 3.1 (0)
10	$\forall P\forall Q(((P \ni (\mathbf{theIdentityFunctionOn}A^{\cdot})) \wedge (Q \ni (\mathbf{theIdentityFunctionOn}A^{\cdot}))) > (((\pi_0 P) = (\pi_0 Q)) > ((\pi_1 P) = (\pi_1 Q))))$	DA 4.8, T 2.9
11	$((\mathbf{theIdentityFunctionOn}A^{\cdot})\mathbf{isAFunction})$	DA 4.1 (1, 10)
100	$((\mathbf{theDomainOf}(\mathbf{theIdentityFunctionOn}A^{\cdot})) = A^{\cdot})$	DA 3.2, DA 4.8, T 2.9, T 1.6 (1)
101	$\forall x((x \ni A^{\cdot}) > ((\mathbf{theValueOf}(\mathbf{theIdentityFunctionOn}A^{\cdot})\mathbf{At}x) = x))$	DA 4.8, DA 4.3, T 2.5 (11, 100)

110	$\forall A(((\mathbf{theIdentityFunctionOn}A)\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}(\mathbf{theIdentityFunctionOn}A)) = A) \wedge \forall x((x \ni A) > ((\mathbf{theValueOf}(\mathbf{theIdentityFunctionOn}A)\mathbf{At}x) = x))))$	$\wedge$I, $\forall$I (11, 100, 101)

$\forall A(((\mathbf{theIdentityFunctionOn}A)\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}(\mathbf{theIdentityFunctionOn}A)) = A) \wedge \forall x((x \ni A) > ((\mathbf{theValueOf}(\mathbf{theIdentityFunctionOn}A)\mathbf{At}x) = x))))$

4.10 The restriction of a function to a set

defining axiom for a function symbol

0	$(\{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\} = \{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\})$	=I
1	$\exists X(X = \{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\})$	$\exists$I (0)
10	$\forall X\forall Y(((X = \{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\}) \wedge (Y = \{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{P \ni F^{\cdot} \mid ((\pi_0 P) \ni A^{\cdot})\})$	D$\exists^{\times}$I (1, 10)

$\forall F\forall A((F\mathbf{isAFunction}) > ((\mathbf{theRestrictionOf}F\mathbf{To}A) = \{P \ni F \mid ((\pi_0 P) \ni A)\}))$

4.11 Equality of functions

theorem

0	$((F^{\cdot}\,\mathbf{isAFunction}) \wedge (G^{\cdot}\,\mathbf{isAFunction}))$	
1	$((F^{\cdot} = G^{\cdot}) > (((\mathbf{theDomainOf}F^{\cdot}) = (\mathbf{theDomainOf}G^{\cdot})) \wedge \forall x((x \ni (\mathbf{theDomainOf}F^{\cdot})) > ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) = (\mathbf{theValueOf}G^{\cdot}\,\mathbf{At}x)))))$	AS, =I, >I
10	$(((\mathbf{theDomainOf}F^{\cdot}) = (\mathbf{theDomainOf}G^{\cdot})) \wedge \forall x((x \ni (\mathbf{theDomainOf}F^{\cdot})) > ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) = (\mathbf{theValueOf}G^{\cdot}\,\mathbf{At}x))))$	
11	$\forall P((P \ni F^{\cdot}) \diamond (P \ni G^{\cdot}))$	DA 4.1, DA 3.2, DA 4.3, T 2.5 (0, 10)
100	$(F^{\cdot} = G^{\cdot})$	T 1.6 (11)
101	$((((\mathbf{theDomainOf}F^{\cdot}) = (\mathbf{theDomainOf}G^{\cdot})) \wedge \forall x((x \ni (\mathbf{theDomainOf}F^{\cdot})) > ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) = (\mathbf{theValueOf}G^{\cdot}\,\mathbf{At}x)))) > (F^{\cdot} = G^{\cdot}))$	>I (10–100)
110	$((F^{\cdot} = G^{\cdot}) \diamond (((\mathbf{theDomainOf}F^{\cdot}) = (\mathbf{theDomainOf}G^{\cdot})) \wedge \forall x((x \ni (\mathbf{theDomainOf}F^{\cdot})) > ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) = (\mathbf{theValueOf}G^{\cdot}\,\mathbf{At}x)))))$	D$\diamond$I (1, 101)
111	$\forall F\forall G(((F\,\mathbf{isAFunction}) \wedge (G\,\mathbf{isAFunction})) > ((F = G) \diamond (((\mathbf{theDomainOf}F) = (\mathbf{theDomainOf}G)) \wedge \forall x((x \ni (\mathbf{theDomainOf}F)) > ((\mathbf{theValueOf}F\,\mathbf{At}x) = (\mathbf{theValueOf}G\,\mathbf{At}x))))))$	>I, $\forall$I (0–110)

$\forall F\forall G(((F\,\mathbf{isAFunction}) \wedge (G\,\mathbf{isAFunction})) > ((F = G) \diamond$

$(((\mathbf{theDomainOf}F) = (\mathbf{theDomainOf}G)) \wedge \forall x((x \ni (\mathbf{theDomainOf}F)) > ((\mathbf{theValueOf}F\,\mathbf{At}x) = (\mathbf{theValueOf}G\,\mathbf{At}x))))))$

4.12 The inverse image of a set under a function

defining axiom for a function symbol

0	$(\{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\} = \{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\})$	=I
1	$\exists X(X = \{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\})$	$\exists$I (0)
10	$\forall X\forall Y(((X = \{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\}) \wedge (Y = \{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\})) > (X = Y))$	AS, =I
11	$\exists^{\times}X(X = \{x \ni (\mathbf{theDomainOf}F^{\cdot}) \mid ((\mathbf{theValueOf}F^{\cdot}\,\mathbf{At}x) \ni B^{\cdot})\})$	D$\exists^{\times}$I (1, 10)

$\forall F\forall B((F\,\mathbf{isAFunction}) > ((\mathbf{theInverseImageOf}B\,\mathbf{Under}F) = \{x \ni (\mathbf{theDomainOf}F) \mid ((\mathbf{theValueOf}F\,\mathbf{At}x) \ni B)\}))$

5 Natural Numbers

5.1 The zero defining axiom for a function symbol

0	$(\bigcirc = \bigcirc)$	=I
1	$\exists X(X = \bigcirc)$	∃I (0)
10	$\forall X \forall Y(((X = \bigcirc) \wedge (Y = \bigcirc)) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \bigcirc)$	D∃$^{\times}$I (1, 10)

$((\mathbf{theZero}) = \bigcirc)$

$(0 = \bigcirc)$

5.2 The successor of a set defining axiom for a function symbol

0	$((A^{\cdot} \cup \{A^{\cdot}\}) = (A^{\cdot} \cup \{A^{\cdot}\}))$	=I
1	$\exists X(X = (A^{\cdot} \cup \{A^{\cdot}\}))$	∃I (0)
10	$\forall X \forall Y(((X = (A^{\cdot} \cup \{A^{\cdot}\})) \wedge (Y = (A^{\cdot} \cup \{A^{\cdot}\}))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (A^{\cdot} \cup \{A^{\cdot}\}))$	D∃$^{\times}$I (1, 10)

$\forall A((A\cdot) > ((\mathbf{theSuccessorOf}\, A) = (A \cup \{A\})))$

5.3 The one defining axiom for a function symbol

0	$((\mathbf{theSuccessorOf}0) = (\mathbf{theSuccessorOf}0))$	=I
1	$\exists X(X = (\mathbf{theSuccessorOf}0))$	∃I (0)
10	$\forall X \forall Y(((X = (\mathbf{theSuccessorOf}0)) \wedge (Y = (\mathbf{theSuccessorOf}0))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theSuccessorOf}0))$	D∃$^{\times}$I (1, 10)

$(1 = (\mathbf{theSuccessorOf}0))$

5.4 The two defining axiom for a function symbol

0	$((\mathbf{theSuccessorOf}1) = (\mathbf{theSuccessorOf}1))$	=I
1	$\exists X(X = (\mathbf{theSuccessorOf}1))$	∃I (0)
10	$\forall X\forall Y(((X = (\mathbf{theSuccessorOf}1)) \wedge (Y = (\mathbf{theSuccessorOf}1))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theSuccessorOf}1))$	D∃×I (1, 10)

$(2 = (\mathbf{theSuccessorOf}1))$

5.5 The elements of zero, one, and two theorem

0	$\forall x((x \ni 1) \diamond (x = 0))$	DA 5.3, DA 5.2, DA 5.1, DA 1.13, DA 1.21, DA 1.18
1	$(1 = \{0\})$	DA 1.18, T 1.6 (0)
10	$\forall x((x \ni 2) \diamond ((x = 0) \vee (x = 1)))$	DA 5.4, DA 5.2, DA 1.21, DA 1.18 (1)
11	$(2 = \{0, 1\})$	DA 1.17, T 1.6 (10)
100	$(0 \ni 1)$	=I, D⋄E (0)
101	$\backslash(0 \ni 0)$	DA 5.1, DA 1.13

110	$\backslash(0 = 1)$	AS, D\I (100, 101)
111	$((1 = \{0\}) \wedge ((2 = \{0, 1\}) \wedge \backslash(0 = 1)))$	∧I (1, 11, 110)

$((1 = \{0\}) \wedge ((2 = \{0, 1\}) \wedge \backslash(0 = 1)))$

5.6 A set being inductive defining axiom for a relation symbol

$\forall A((A\cdot) > ((A\mathbf{isInductive}) \diamond ((0 \ni A) \wedge \forall x((x \ni A) > ((\mathbf{theSuccessorOf}x) \ni A)))))$

5.7 The axiom of infinity axiom

$\exists I((I\cdot) \wedge (I\textbf{isInductive}))$

5.8 The set of the natural numbers defining axiom for a function symbol

0	$\exists I(I\textbf{isInductive})$	A 5.7, ∃E, ∃I
1	$(I^{\times}\textbf{isInductive})$	
10	$\exists N\forall x((x \ni N) \diamond ((x \ni I^{\times}) \wedge \forall J((J\textbf{isInductive}) > (x \ni J))))$	AS 1.14
11	$\forall x(\forall J((J\textbf{isInductive}) > (x \ni J)) > (x \ni I^{\times}))$	∀E, >E, >I, ∀I (1)
100	$\exists N\forall x((x \ni N) \diamond \forall J((J\textbf{isInductive}) > (x \ni J)))$	∃E, ∧E, ∧I, D◇I, ∃I (10, 11)
101	$\exists N\forall x((x \ni N) \diamond \forall J((J\textbf{isInductive}) > (x \ni J)))$	∃E (0, 1–100)

110	$\exists^{\times} N\forall x((x \ni N) \diamond \forall J((J\textbf{isInductive}) > (x \ni J)))$	TS 1.7 (101)

$\forall x((x\cdot) > ((x \ni (\textbf{theSetOfTheNaturalNumbers})) \diamond \forall I((I\textbf{isInductive}) > (x \ni I))))$

$\forall x((x\cdot) > ((x \ni \mathbb{N}) \diamond \forall I((I\textbf{isInductive}) > (x \ni I))))$

5.9 A set being a natural number defining axiom for a relation symbol

$\forall n((n\cdot) > ((n\textbf{isANaturalNumber}) \diamond (n \ni \mathbb{N})))$

5.10 The natural numbers form an inductive set theorem

0	$\forall I((I\textbf{isInductive}) > (0 \ni I))$	DA 5.6, ∧E
1	$(0 \ni \mathbb{N})$	DA 5.8 (0)
10	$(n' \ni \mathbb{N})$	
11	$\forall I((I\textbf{isInductive}) > (n' \ni I))$	DA 5.8 (10)
100	$\forall I((I\textbf{isInductive}) > ((\textbf{theSuccessorOf}n') \ni I))$	DA 5.6, ∀E, >E, >I, ∀I (11)
101	$((\textbf{theSuccessorOf}n') \ni \mathbb{N})$	DA 5.8 (100)

110	$\forall n((n \ni \mathbb{N}) > ((\mathbf{theSuccessorOf}\, n) \ni \mathbb{N}))$	>I, ∀I (10–101)
111	$(\mathbb{N}\mathbf{isInductive})$	DA 5.6, ∧I (1, 110)

$(\mathbb{N}\mathbf{isInductive})$

5.11 The natural numbers form the least inductive set theorem

0	$(I\mathbf{isInductive})$	
1	$\forall n((n \ni \mathbb{N}) > (n \ni I))$	DA 5.8, ∀E, >E, >I, ∀I (0)
10	$(\mathbb{N} \supset I)$	DA 1.8 (1)
11	$\forall I((I\mathbf{isInductive}) > (\mathbb{N} \supset I))$	>I, ∀I (0–10)

$\forall I((I\mathbf{isInductive}) > (\mathbb{N} \supset I))$

5.12 The principle of induction theorem schema

For each formula $\phi[n]$, with arbitrary parameters. Substitutions avoid variable capture.

0	$(\phi[0] \wedge \forall n((n \ni \mathbb{N}) > (\phi[n] > \phi[(\mathbf{theSuccessorOf}\, n)])))$	
1	$(0 \ni \{n \ni \mathbb{N} \mid \phi[n]\})$	T 5.10, DA 5.6, DA 1.15, ∧E (0)
10	$\forall n((n \ni \{k \ni \mathbb{N} \mid \phi[k]\}) > ((\mathbf{theSuccessorOf}\, n) \ni \{k \ni \mathbb{N} \mid \phi[k]\}))$	T 5.10, DA 5.6, DA 1.15, ∀E, >E (0)
11	$(\{n \ni \mathbb{N} \mid \phi[n]\}\mathbf{isInductive})$	DA 5.6, ∧I (1, 10)
100	$(\mathbb{N} \supset \{n \ni \mathbb{N} \mid \phi[n]\})$	T 5.11 (11)
101	$\forall n((n \ni \mathbb{N}) > \phi[n])$	DA 1.8, DA 1.15 (100)

110	$((\phi[0] \wedge \forall n((n \ni \mathbb{N}) > (\phi[n] > \phi[(\mathbf{theSuccessorOf}\, n)]))) > \forall n((n \ni \mathbb{N}) > \phi[n]))$	>I (0–101)

$((\phi[0] \wedge \forall n((n \ni \mathbb{N}) > (\phi[n] > \phi[(\mathbf{theSuccessorOf}\, n)]))) > \forall n((n \ni \mathbb{N}) > \phi[n]))$

5.13 A natural number being less than a natural number defining axiom for a relation symbol

$\forall m((m \ni \mathbb{N}) > \forall n((n \ni \mathbb{N}) > ((m \mathbf{isLessThan} n) \diamond (m \ni n))))$

$\forall m((m \ni \mathbb{N}) > \forall n((n \ni \mathbb{N}) > ((m < n) \diamond (m \ni n))))$

5.14 Zero, one, and two are natural numbers theorem

0	$(0 \ni \mathbb{N})$	T 5.10, DA 5.6
1	$(1 \ni \mathbb{N})$	T 5.10, DA 5.6, DA 5.3 (0)
10	$(2 \ni \mathbb{N})$	T 5.10, DA 5.6, DA 5.4 (1)
11	$((0 \ni \mathbb{N}) \wedge ((1 \ni \mathbb{N}) \wedge (2 \ni \mathbb{N})))$	$\wedge$I (0, 1, 10)

$((0 \ni \mathbb{N}) \wedge ((1 \ni \mathbb{N}) \wedge (2 \ni \mathbb{N})))$

6 Tuples

6.1 The tuple of two sets defining axiom for a function symbol

0	$(\{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\} = \{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\})$	=I
1	$\exists X(X = \{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\})$	∃I (0)
10	$\forall X \forall Y(((X = \{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\}) \wedge (Y = \{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{\langle 0, A^{\cdot}\rangle, \langle 1, B^{\cdot}\rangle\})$	D∃$^{\times}$I (1, 10)

$\forall A \forall B((\mathbf{theTupleOf}\, A \mathbf{And} B) = \{\langle 0, A\rangle, \langle 1, B\rangle\})$

$\forall A \forall B([A, B] = \{\langle 0, A\rangle, \langle 1, B\rangle\})$

6.2 A tuple of two sets is a function on two theorem

0	$\forall P((P \ni [A^{\cdot}, B^{\cdot}]) \diamond ((P = \langle 0, A^{\cdot}\rangle) \vee (P = \langle 1, B^{\cdot}\rangle)))$	DA 6.1, DA 1.17
1	$([A^{\cdot}, B^{\cdot}]\mathbf{isARelation})$	DA 2.6, DA 3.1, ∨E (0)
10	$\forall P \forall Q(((P \ni [A^{\cdot}, B^{\cdot}]) \wedge (Q \ni [A^{\cdot}, B^{\cdot}])) > (((\pi_0 P) = (\pi_0 Q)) > ((\pi_1 P) = (\pi_1 Q))))$	T 2.9, T 5.5, ∨E (0)
11	$([A^{\cdot}, B^{\cdot}]\mathbf{isAFunction})$	DA 4.1 (1, 10)
100	$\forall x((x \ni (\mathbf{theDomainOf}[A^{\cdot}, B^{\cdot}])) \diamond ((x = 0) \vee (x = 1)))$	DA 3.2, T 2.9 (0, 1)
101	$((\mathbf{theDomainOf}[A^{\cdot}, B^{\cdot}]) = 2)$	T 5.5, DA 1.17, T 1.6 (100)

110	$\forall A \forall B(([A, B]\mathbf{isAFunction}) \wedge ((\mathbf{theDomainOf}[A, B]) = 2))$	∧I, ∀I (11, 101)

$\forall A \forall B(([A, B]\mathbf{isAFunction}) \wedge ((\mathbf{theDomainOf}[A, B]) = 2))$

6.3 The components of a tuple of two sets theorem

0	$([A^{\cdot}, B^{\cdot}]\mathbf{isAFunction})$	T 6.2, ∧E
1	$((\langle 0, A^{\cdot}\rangle \ni [A^{\cdot}, B^{\cdot}]) \wedge (\langle 1, B^{\cdot}\rangle \ni [A^{\cdot}, B^{\cdot}]))$	DA 6.1, DA 1.17
10	$(((\mathbf{theValueOf}[A^{\cdot}, B^{\cdot}]\mathbf{At}0) = A^{\cdot}) \wedge ((\mathbf{theValueOf}[A^{\cdot}, B^{\cdot}]\mathbf{At}1) = B^{\cdot}))$	DA 4.3, DA 4.1, T 2.9 (0, 1)
11	$\forall A \forall B(((\mathbf{theValueOf}[A, B]\mathbf{At}0) = A) \wedge ((\mathbf{theValueOf}[A, B]\mathbf{At}1) = B))$	∀I (10)

$\forall A \forall B(((\mathbf{theValueOf}[A, B]\mathbf{At}0) = A) \wedge ((\mathbf{theValueOf}[A, B]\mathbf{At}1) = B))$

6.4 A set being a tuple

defining axiom for a relation symbol

$\forall T((T\cdot) > ((T\mathbf{isATuple}) \diamond ((T\mathbf{isAFunction}) \wedge \exists n((n \ni \mathbb{N}) \wedge ((\mathbf{theDomainOf}T) = n)))))$

6.5 The length of a tuple

defining axiom for a function symbol

0	$((\mathbf{theDomainOf}T^{\cdot}) = (\mathbf{theDomainOf}T^{\cdot}))$	=I
1	$\exists X(X = (\mathbf{theDomainOf}T^{\cdot}))$	∃I (0)
10	$\forall X \forall Y(((X = (\mathbf{theDomainOf}T^{\cdot})) \wedge (Y = (\mathbf{theDomainOf}T^{\cdot}))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theDomainOf}T^{\cdot}))$	D∃×I (1, 10)

$\forall T((T\mathbf{isATuple}) > ((\mathbf{theLengthOf}T) = (\mathbf{theDomainOf}T)))$

6.6 The component of a tuple at a natural number

defining axiom for a function symbol

0	$((\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot}) = (\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot}))$	=I
1	$\exists X(X = (\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot}))$	∃I (0)
10	$\forall X \forall Y(((X = (\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot})) \wedge (Y = (\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot}))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}T^{\cdot}\mathbf{At}i^{\cdot}))$	D∃×I (1, 10)

$\forall T \forall i(((T\mathbf{isATuple}) \wedge (i \ni (\mathbf{theDomainOf}T))) > ((\mathbf{theComponentOf}T\mathbf{At}i) = (\mathbf{theValueOf}T\mathbf{At}i)))$

$\forall T \forall i(((T\mathbf{isATuple}) \wedge (i \ni (\mathbf{theDomainOf}T))) > ((Ti) = (\mathbf{theValueOf}T\mathbf{At}i)))$

6.7 The empty tuple defining axiom for a function symbol

0	$(\bigcirc = \bigcirc)$	=I
1	$\exists X(X = \bigcirc)$	∃I (0)
10	$\forall X \forall Y(((X = \bigcirc) \wedge (Y = \bigcirc)) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \bigcirc)$	D∃$^{\times}$I (1, 10)

$((\mathbf{theEmptyTuple}) = \bigcirc)$

6.8 The empty set is a tuple of length zero theorem

0	$(\bigcirc\mathbf{isARelation})$	DA 1.13, DA 3.1
1	$(\bigcirc\mathbf{isAFunction})$	DA 1.13, DA 4.1 (0)
10	$((\mathbf{theDomainOf}\bigcirc) = 0)$	DA 3.2, DA 1.13, DA 5.1, T 1.6 (0)
11	$(0 \ni \mathbb{N})$	T 5.10, DA 5.6
100	$(\bigcirc\mathbf{isATuple})$	DA 6.4, ∃I (1, 10, 11)
101	$((\bigcirc\mathbf{isATuple}) \wedge ((\mathbf{theLengthOf}\bigcirc) = 0))$	DA 6.5, ∧I (10, 100)

$((\bigcirc\mathbf{isATuple}) \wedge ((\mathbf{theLengthOf}\bigcirc) = 0))$

7 Relation (continued)

7.1 The Cartesian product of two sets defining axiom for a function symbol

0	$\forall x \forall y (((x \ni A^{\cdot}) \wedge (y \ni B^{\cdot})) > (\langle x, y \rangle \ni \mathcal{P}(\mathcal{P}(A^{\cdot} \cup B^{\cdot}))))$	T 2.10, DA 1.21
1	$\exists C \forall P ((P \ni C) \diamond \exists x \exists y (((x \ni A^{\cdot}) \wedge (y \ni B^{\cdot})) \wedge (P = \langle x, y \rangle)))$	AS 1.14 (0)
10	$\forall A \forall B \exists^{\times} C \forall P ((P \ni C) \diamond \exists x \exists y (((x \ni A) \wedge (y \ni B)) \wedge (P = \langle x, y \rangle)))$	TS 1.7, ∀I (1)

$\forall A \forall B \forall P ((P \ni (\mathbf{theCartesianProductOf}\, A\, \mathbf{And}\, B)) \diamond \exists x \exists y (((x \ni A) \wedge (y \ni B)) \wedge (P = \langle x, y \rangle)))$

$\forall A \forall B \forall P ((P \ni (A \times B)) \diamond \exists x \exists y (((x \ni A) \wedge (y \ni B)) \wedge (P = \langle x, y \rangle)))$

7.2 Membership of a Cartesian product theorem

0	$\forall x \forall y ((\langle x, y \rangle \ni (A^{\cdot} \times B^{\cdot})) \diamond \exists a \exists b (((a \ni A^{\cdot}) \wedge (b \ni B^{\cdot})) \wedge (\langle x, y \rangle = \langle a, b \rangle)))$	DA 7.1
1	$\forall x \forall y ((\langle x, y \rangle \ni (A^{\cdot} \times B^{\cdot})) \diamond ((x \ni A^{\cdot}) \wedge (y \ni B^{\cdot})))$	T 2.5, ∃E, ∃I (0)
10	$\forall A \forall B \forall x \forall y ((\langle x, y \rangle \ni (A \times B)) \diamond ((x \ni A) \wedge (y \ni B)))$	∀I (1)

$\forall A \forall B \forall x \forall y ((\langle x, y \rangle \ni (A \times B)) \diamond ((x \ni A) \wedge (y \ni B)))$

7.3 A Cartesian product is a relation theorem

0	$\forall P ((P \ni (A^{\cdot} \times B^{\cdot})) > (P\mathbf{isAPair}))$	DA 7.1, DA 2.6, ∃E
1	$\forall A \forall B ((A \times B)\mathbf{isARelation})$	DA 3.1, ∀I (0)

$\forall A \forall B ((A \times B)\mathbf{isARelation})$

7.4 A Cartesian product with the empty set theorem

0	$\forall P \backslash (P \ni (A^{\cdot} \times \bigcirc))$	DA 7.1, DA 1.13, ∃E, D\I
1	$\forall P \backslash (P \ni (\bigcirc \times A^{\cdot}))$	DA 7.1, DA 1.13, ∃E, D\I
10	$\forall A(((A \times \bigcirc) = \bigcirc) \wedge ((\bigcirc \times A) = \bigcirc))$	T 1.29, ∧I, ∀I (0, 1)

$\forall A(((A \times \bigcirc) = \bigcirc) \wedge ((\bigcirc \times A) = \bigcirc))$

7.5 The set of tuples of a given length in a set defining axiom for a function symbol

0	$\forall T(((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n^{\cdot}) \wedge ((\mathbf{theRangeOf}T) \supset A^{\cdot}))) > (T \ni \mathcal{P}((n^{\cdot} \times A^{\cdot}))))$	DA 4.1, DA 3.2, DA 3.3, DA 1.25, T 7.2
1	$\exists S \forall T((T \ni S) \diamond ((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n^{\cdot}) \wedge ((\mathbf{theRangeOf}T) \supset A^{\cdot}))))$	AS 1.14 (0)
10	$\forall n \forall A \exists^{\times} S \forall T((T \ni S) \diamond ((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A))))$	TS 1.7, ∀I (1)

$\forall n((n \ni \mathbb{N}) > \forall A \forall T((T \ni (\mathbf{theSetOfTuplesOfLength}n\mathbf{In}A)) \diamond ((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A)))))$

$\forall n((n \ni \mathbb{N}) > \forall A \forall T((T \ni A^n) \diamond ((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A)))))$

7.6 Membership of a tuple power theorem

0	$\forall n((n \ni \mathbb{N}) > \forall A \forall T((T \ni A^n) \diamond ((T\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A)))))$	DA 7.5
1	$\forall n((n \ni \mathbb{N}) > \forall A \forall T((T \ni A^n) \diamond ((T\mathbf{isATuple}) \wedge (((\mathbf{theLengthOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A)))))$	DA 6.4, DA 6.5 (0)

$\forall n((n \ni \mathbb{N}) > \forall A \forall T((T \ni A^n) \diamond ((T\mathbf{isATuple}) \wedge (((\mathbf{theLengthOf}T) = n) \wedge ((\mathbf{theRangeOf}T) \supset A)))))$

7.7 A binary tuple in a tuple power theorem

0	$\forall x \forall y(([x,y]\mathbf{isAFunction}) \wedge ((\mathbf{theDomainOf}[x,y]) = 2))$	T 6.2
1	$\forall x \forall y((\mathbf{theRangeOf}[x,y]) = \{x,y\})$	DA 6.1, DA 3.3, T 2.9, T 1.6
10	$\forall A \forall x \forall y(([x,y] \ni A^2) \diamond ((x \ni A) \wedge (y \ni A)))$	T 5.14, DA 7.5, DA 1.8, DA 1.17 (0, 1)

$\forall A \forall x \forall y (([x, y] \ni A^2) \diamond ((x \ni A) \wedge (y \ni A)))$

7.8 The zeroth tuple power theorem

0	$\forall T ((T \ni A^{\cdot 0}) \diamond ((T \mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf} T) = 0) \wedge ((\mathbf{theRangeOf} T) \supset A^{\cdot}))))$	T 5.14, DA 7.5
1	$\forall T (((T \mathbf{isAFunction}) \wedge ((\mathbf{theDomainOf} T) = 0)) > (T = \bigcirc))$	DA 4.1, DA 3.2, DA 5.1, DA 1.13, T 1.29
10	$\forall T ((T \ni A^{\cdot 0}) \diamond (T = \bigcirc))$	T 6.8, DA 3.3, DA 1.8 (0, 1)
11	$\forall A (A^0 = \{\bigcirc\})$	DA 1.18, T 1.6, ∀I (10)

$\forall A (A^0 = \{\bigcirc\})$

7.9 A set being an n-ary relation on a set defining axiom for a relation symbol

$\forall n ((n \ni \mathbb{N}) > \forall R \forall A ((R \mathbf{isAnNAryRelationOfArity} n \mathbf{On} A) \diamond (R \supset A^n)))$

7.10 Nullary relations on a set theorem

0	$\forall A \forall R ((R \mathbf{isAnNAryRelationOfArity} 0 \mathbf{On} A) \diamond (R \supset \{\bigcirc\}))$	DA 7.9, T 5.14, T 7.8
1	$\forall R ((R \supset \{\bigcirc\}) \diamond ((R = \bigcirc) \vee (R = \{\bigcirc\})))$	DA 1.8, DA 1.18, T 1.6, T 1.29, D\E, ∨E
10	$\forall A \forall R ((R \mathbf{isAnNAryRelationOfArity} 0 \mathbf{On} A) \diamond ((R = \bigcirc) \vee (R = \{\bigcirc\})))$	D⋄E, D⋄I (0, 1)

$\forall A \forall R ((R \mathbf{isAnNAryRelationOfArity} 0 \mathbf{On} A) \diamond ((R = \bigcirc) \vee (R = \{\bigcirc\})))$

7.11 The tuple relation associated with a relation defining axiom for a function symbol

0	$(\{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\} = \{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\})$	=I
1	$\exists X(X = \{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\})$	∃I (0)
10	$\forall X \forall Y(((X = \{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\}) \wedge (Y = \{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{T \ni A^{\cdot 2} \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R^{\cdot})\})$	D∃×I (1, 10)

$\forall R \forall A((R\mathbf{isARelationOn}A) > ((\mathbf{theTupleRelationAssociatedWith}R\mathbf{On}A) = \{T \ni A^2 \mid (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R)\}))$

7.12 Membership of the associated tuple relation theorem

0	$\forall R \forall A((R\mathbf{isARelationOn}A) > \forall T((T \ni (\mathbf{theTupleRelationAssociatedWith}R\mathbf{On}A)) \diamond ((T \ni A^2) \wedge (\langle(\mathbf{theValueOf}T\mathbf{At}0), (\mathbf{theValueOf}T\mathbf{At}1)\rangle \ni R))))$	DA 7.11, DA 1.15
1	$\forall R \forall A((R\mathbf{isARelationOn}A) > \forall x((x \ni A) > \forall y((y \ni A) > (([x, y] \ni (\mathbf{theTupleRelationAssociatedWith}R\mathbf{On}A)) \diamond (\langle x, y\rangle \ni R)))))$	T 7.7, T 6.3 (0)

$\forall R \forall A((R\mathbf{isARelationOn}A) > \forall x((x \ni A) > \forall y((y \ni A) > (([x, y] \ni (\mathbf{theTupleRelationAssociatedWith}R\mathbf{On}A)) \diamond (\langle x, y\rangle \ni R)))))$

8 Function (continued)

8.1 The set of functions from a set to a set defining axiom for a function symbol

0	$(\{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\} = \{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\})$	=I
1	$\exists X(X = \{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\})$	∃I (0)
10	$\forall X\forall Y(((X = \{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\}) \wedge (Y = \{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{F \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})) \mid (F\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})\})$	D∃×I (1, 10)

$\forall A\forall B((\mathbf{theSetOfFunctionsFrom}A\mathbf{To}B) = \{F \ni \mathcal{P}((A \times B)) \mid (F\mathbf{isAFunctionFrom}A\mathbf{To}B)\})$

8.2 Membership of the set of functions theorem

0	$(F^{\cdot}\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot})$	
1	$\forall P((P \ni F^{\cdot}) > (P \ni (A^{\cdot} \times B^{\cdot})))$	DA 4.4, DA 3.1, DA 3.2, DA 3.3, T 7.2 (0)
10	$(F^{\cdot} \ni \mathcal{P}((A^{\cdot} \times B^{\cdot})))$	DA 1.8, DA 1.25 (1)
11	$(F^{\cdot} \ni (\mathbf{theSetOfFunctionsFrom}A^{\cdot}\mathbf{To}B^{\cdot}))$	DA 8.1, DA 1.15 (0, 10)
100	$((F^{\cdot}\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot}) > (F^{\cdot} \ni (\mathbf{theSetOfFunctionsFrom}A^{\cdot}\mathbf{To}B^{\cdot})))$	>I (0–11)
101	$((F^{\cdot} \ni (\mathbf{theSetOfFunctionsFrom}A^{\cdot}\mathbf{To}B^{\cdot})) > (F^{\cdot}\mathbf{isAFunctionFrom}A^{\cdot}\mathbf{To}B^{\cdot}))$	DA 8.1, DA 1.15, ∧E

110	$\forall A\forall B\forall F((F \ni (\mathbf{theSetOfFunctionsFrom}A\mathbf{To}B)) \diamond (F\mathbf{isAFunctionFrom}A\mathbf{To}B))$	D◇I, ∀I (100, 101)

$\forall A\forall B\forall F((F \ni (\mathbf{theSetOfFunctionsFrom}A\mathbf{To}B)) \diamond (F\mathbf{isAFunctionFrom}A\mathbf{To}B))$

8.3 A function being an n-ary function defining axiom for a relation symbol

$\forall n((n \ni \mathbb{N}) > \forall F\forall A\forall B((F\mathbf{isAnNAryFunctionOfArity}n\mathbf{From}A\mathbf{To}B) \diamond (F\mathbf{isAFunctionFrom}A^{n}\mathbf{To}B)))$

8.4 The value of a binary function at two sets defining axiom for a function symbol

0	$((\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}]) = (\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}]))$	=I
1	$\exists X(X = (\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}]))$	∃I (0)
10	$\forall X\forall Y(((X = (\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}])) \wedge (Y = (\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}]))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}F^{\cdot}\mathbf{At}[x^{\cdot},y^{\cdot}]))$	D∃×I (1, 10)

$\forall F\forall A\forall B((F\mathbf{isAnNAryFunctionOfArity}2\mathbf{From}A\mathbf{To}B) > \forall x((x \ni A) > \forall y((y \ni A) > ((\mathbf{theBinaryValueOf}F\mathbf{At}x\mathbf{And}y) = (\mathbf{theValueOf}F\mathbf{At}[x,y])))))$

$\forall F\forall A\forall B((F\mathbf{isAnNAryFunctionOfArity}2\mathbf{From}A\mathbf{To}B) > \forall x((x \ni A) > \forall y((y \ni A) > ((xFy) = (\mathbf{theValueOf}F\mathbf{At}[x,y])))))$

8.5 A function being an n-ary operation on a set defining axiom for a relation symbol

$\forall n((n \ni \mathbb{N}) > \forall F\forall A((F\mathbf{isAnNAryOperationOfArity}n\mathbf{On}A) \diamond (F\mathbf{isAnNAryFunctionOfArity}n\mathbf{From}A\mathbf{To}A)))$

8.6 A function being a binary operation on a set defining axiom for a relation symbol

$\forall F\forall A((F\mathbf{isABinaryOperationOn}A) \diamond (F\mathbf{isAnNAryFunctionOfArity}2\mathbf{From}A\mathbf{To}A))$

8.7 A function being a unary operation on a set defining axiom for a relation symbol

$\forall F\forall A((F\mathbf{isAUnaryOperationOn}A) \diamond (F\mathbf{isAnNAryFunctionOfArity}1\mathbf{From}A\mathbf{To}A))$

8.8 Values of an n-ary function theorem

0	$((n^{\cdot} \ni \mathbb{N}) \wedge (F^{\cdot}\mathbf{isAnNAryFunctionOfArity}n^{\cdot}\mathbf{From}A^{\cdot}\mathbf{To}B^{\cdot}))$	
1	$((F^{\cdot}\mathbf{isAFunction}) \wedge (((\mathbf{theDomainOf}F^{\cdot}) = A^{\cdot n^{\cdot}}) \wedge ((\mathbf{theRangeOf}F^{\cdot}) \supset B^{\cdot})))$	DA 8.3, DA 4.4 (0)
10	$\forall T((T \ni A^{\cdot n^{\cdot}}) > (\langle T, (\mathbf{theValueOf}F^{\cdot}\mathbf{At}T)\rangle \ni F^{\cdot}))$	DA 4.3 (1)
11	$\forall T((T \ni A^{\cdot n^{\cdot}}) > ((\mathbf{theValueOf}F^{\cdot}\mathbf{At}T) \ni B^{\cdot}))$	DA 3.3, T 2.9, DA 1.8 (1, 10)
100	$\forall n((n \ni \mathbb{N}) > \forall F\forall A\forall B((F\mathbf{isAnNAryFunctionOfArity}n\mathbf{From}A\mathbf{To}B) > \forall T((T \ni A^n) > ((\mathbf{theValueOf}F\mathbf{At}T) \ni B))))$	>I, ∀I (0–11)

$\forall n((n \ni \mathbb{N}) > \forall F \forall A \forall B((F \textbf{isAnNAryFunctionOfArity} n \textbf{FromA} A \textbf{ToB} B) > \forall T((T \ni A^n) > ((\textbf{theValueOf} F \textbf{At} T) \ni B))))$

8.9 Closure of a binary operation theorem

0	$\forall F \forall A((F \textbf{isABinaryOperationOn} A) > (F \textbf{isAnNAryFunctionOfArity} 2 \textbf{FromA} A \textbf{To} A))$	DA 8.6
1	$\forall F \forall A((F \textbf{isABinaryOperationOn} A) > \forall x((x \ni A) > \forall y((y \ni A) > ((\textbf{theValueOf} F \textbf{At} [x, y]) \ni A))))$	T 8.8, T 5.14, T 7.7 (0)
10	$\forall F \forall A((F \textbf{isABinaryOperationOn} A) > \forall x((x \ni A) > \forall y((y \ni A) > ((x\, F\, y) \ni A))))$	DA 8.4 (1)

$\forall F \forall A((F \textbf{isABinaryOperationOn} A) > \forall x((x \ni A) > \forall y((y \ni A) > ((x\, F\, y) \ni A))))$

2 Algebra

2.1 Basic Algebra

1 Magmas, Semigroups, and Monoids

1.1 An object being a magma defining axiom for a relation symbol

$$\forall M((M\,\mathbf{isAMagma}) \diamond ((M\,\mathbf{isATuple}) \wedge (((\mathbf{theLengthOf}\,M) = 2) \wedge ((\mathbf{theValueOf}\,M\,\mathbf{At}1)\mathbf{isABinaryOperationOn}(\mathbf{theValueOf}\,M\,\mathbf{At}0)))))$$

1.2 The underlying set of a magma defining axiom for a function symbol

0	$((\mathbf{theValueOf}\,M\,\mathbf{At}0) = (\mathbf{theValueOf}\,M\,\mathbf{At}0))$	=I
1	$\exists X(X = (\mathbf{theValueOf}\,M\,\mathbf{At}0))$	∃I (0)
10	$\forall X\forall Y(((X = (\mathbf{theValueOf}\,M\,\mathbf{At}0)) \wedge (Y = (\mathbf{theValueOf}\,M\,\mathbf{At}0))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}\,M\,\mathbf{At}0))$	D∃×I (1, 10)

$$\forall M((M\,\mathbf{isAMagma}) > ((\mathbf{theUnderlyingSetOf}\,M) = (\mathbf{theValueOf}\,M\,\mathbf{At}0)))$$

1.3 The binary operation of a magma defining axiom for a function symbol

0	$((\mathbf{theValueOf}\,M\,\mathbf{At}1) = (\mathbf{theValueOf}\,M\,\mathbf{At}1))$	=I
1	$\exists X(X = (\mathbf{theValueOf}\,M\,\mathbf{At}1))$	∃I (0)
10	$\forall X\forall Y(((X = (\mathbf{theValueOf}\,M\,\mathbf{At}1)) \wedge (Y = (\mathbf{theValueOf}\,M\,\mathbf{At}1))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}\,M\,\mathbf{At}1))$	D∃×I (1, 10)

$$\forall M((M\,\mathbf{isAMagma}) > ((\mathbf{theBinaryOperationOf}\,M) = (\mathbf{theValueOf}\,M\,\mathbf{At}1)))$$

1.4 A binary operation being associative on a set defining axiom for a relation symbol

$$\forall F\forall A((F\mathbf{isABinaryOperationOn}A) > ((F\mathbf{isAssociativeOn}A) \diamond \forall x((x \ni A) > \forall y((y \ni A) > \forall z((z \ni A) > (((x\,F\,y)\,F\,z) = (x\,F\,(y\,F\,z))))))))$$

1.5 An object being an identity element defining axiom for a relation symbol

$\forall F \forall A \forall e((e \textbf{isAnIdentityElementWithRespectTo} F \textbf{Of} A) \diamond ((e \ni A) \wedge \forall x((x \ni A) > (((e\, F\, x) = x) \wedge ((x\, F\, e) = x)))))$

1.6 An object being an inverse element defining axiom for a relation symbol

$\forall F \forall A \forall e \forall a \forall b((b \textbf{isAnInverseElementOf} a \textbf{WithRespectTo} e \textbf{Under} F \textbf{Of} A) \diamond ((b \ni A) \wedge ((a \ni A) \wedge ((e \ni A) \wedge (((b\, F\, a) = e) \wedge ((a\, F\, b) = e))))))$

1.7 Uniqueness of an identity element theorem

0	$((e^{\cdot} \textbf{isAnIdentityElementWithRespectTo} F^{\cdot} \textbf{Of} A^{\cdot}) \wedge (f^{\cdot} \textbf{isAnIdentityElementWithRespectTo} F^{\cdot} \textbf{Of} A^{\cdot}))$	
1	$((e^{\cdot} \ni A^{\cdot}) \wedge (f^{\cdot} \ni A^{\cdot}))$	DA BA 1.5, ∧E (0)
10	$((e^{\cdot}\, F^{\cdot}\, f^{\cdot}) = f^{\cdot})$	DA BA 1.5, ∀E, >E (0, 1)
11	$((e^{\cdot}\, F^{\cdot}\, f^{\cdot}) = e^{\cdot})$	DA BA 1.5, ∀E, >E (0, 1)
100	$(e^{\cdot} = f^{\cdot})$	AS (10, 11)
101	$\forall F \forall A \forall e \forall f(((e \textbf{isAnIdentityElementWithRespectTo} F \textbf{Of} A) \wedge (f \textbf{isAnIdentityElementWithRespectTo} F \textbf{Of} A)) > (e = f))$	>I, ∀I (0–100)

$\forall F \forall A \forall e \forall f(((e \textbf{isAnIdentityElementWithRespectTo} F \textbf{Of} A) \wedge (f \textbf{isAnIdentityElementWithRespectTo} F \textbf{Of} A)) > (e = f))$

1.8 A magma being a semigroup defining axiom for a relation symbol

$\forall M((M \textbf{isASemigroup}) \diamond ((M \textbf{isAMagma}) \wedge ((\textbf{theBinaryOperationOf} M) \textbf{isAssociativeOn} (\textbf{theUnderlyingSetOf} M))))$

1.9 A magma being a monoid defining axiom for a relation symbol

$\forall M((M \textbf{isAMonoid}) \diamond ((M \textbf{isASemigroup}) \wedge \exists e(e \textbf{isAnIdentityElementWithRespectTo} (\textbf{theBinaryOperationOf} M) \textbf{Of} (\textbf{theUnderlyingSetOf} M))))$

1.10 The identity element of a monoid defining axiom for a function symbol

0	$\forall M((M\mathbf{isAMonoid}) > \exists e(e\mathbf{isAnIdentityElementWithRespectTo}(\mathbf{theBinaryOperationOf}M)\mathbf{Of}(\mathbf{theUnderlyingSetOf}M)))$	DA BA 1.9
1	$\forall M\forall e\forall f(((e\mathbf{isAnIdentityElementWithRespectTo}(\mathbf{theBinaryOperationOf}M)\mathbf{Of}(\mathbf{theUnderlyingSetOf}M)) \wedge (f\mathbf{isAnIdentityElementWithRespectTo}(\mathbf{theBinaryOperationOf}M)\mathbf{Of}(\mathbf{theUnderlyingSetOf}M))) > (e = f))$	T BA 1.7
10	$\forall M((M\mathbf{isAMonoid}) > \exists^{\times} e(e\mathbf{isAnIdentityElementWithRespectTo}(\mathbf{theBinaryOperationOf}M)\mathbf{Of}(\mathbf{theUnderlyingSetOf}M)))$	D$\exists^{\times}$I (0, 1)

$$\forall M((M\mathbf{isAMonoid}) > ((\mathbf{theIdentityElementOf}M)\mathbf{isAnIdentityElementWithRespectTo}(\mathbf{theBinaryOperationOf}M)\mathbf{Of}(\mathbf{theUnderlyingSetOf}M)))$$

1 Groups

1.1 A magma being a group defining axiom for a relation symbol

$\forall G((G\mathbf{isAGroup})\diamond((G\mathbf{isAMonoid})\wedge\forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > \exists b(b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOpera}$
$G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G)))))$

1.2 A group is a magma theorem

0	$(G\,\mathbf{isAGroup})$	
1	$(G\,\mathbf{isAMonoid})$	DA GT 1.1, ∧E (0)
10	$(G\,\mathbf{isASemigroup})$	DA BA 1.9, ∧E (1)
11	$(G\,\mathbf{isAMagma})$	DA BA 1.8, ∧E (10)
100	$\forall G((G\mathbf{isAGroup}) > (G\mathbf{isAMagma}))$	>I, ∀I (0–11)

$\forall G((G\mathbf{isAGroup}) > (G\mathbf{isAMagma}))$

1.3 Uniqueness of an inverse in a group theorem

0	$((G\,\mathbf{isAGroup}) \wedge ((b\,\mathbf{isAnInverseElementOf}a\,\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}$ $(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G)) \wedge (c\,\mathbf{isAnInverseElementOf}a\,\mathbf{WithRespectTo}$ $(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))))$	
1	$((a \ni (\mathbf{theUnderlyingSetOf}G)) \wedge ((b \ni (\mathbf{theUnderlyingSetOf}G)) \wedge (c \ni (\mathbf{theUnderlyingSetOf}G))))$	DA BA 1.6 (0)
10	$(b = (b\,(\mathbf{theBinaryOperationOf}G)\,(\mathbf{theIdentityElementOf}G)))$	DA BA 1.10, DA BA 1.5, DA GT 1.1 (0, 1)
11	$((b\,(\mathbf{theBinaryOperationOf}G)\,(\mathbf{theIdentityElementOf}G)) = (b\,(\mathbf{theBinaryOperationOf}$ $G)\,(a\,(\mathbf{theBinaryOperationOf}G)\,c)))$	DA BA 1.6, AS (0)
100	$((b\,(\mathbf{theBinaryOperationOf}G)\,(a\,(\mathbf{theBinaryOperationOf}G)\,c)) = ((b\,(\mathbf{theBinaryOperationOf}$ $G)\,a)\,(\mathbf{theBinaryOperationOf}G)\,c))$	DA BA 1.4, DA BA 1.8, DA BA 1.9, DA GT 1.1 (0, 1)
101	$(((b\,(\mathbf{theBinaryOperationOf}G)\,a)\,(\mathbf{theBinaryOperationOf}G)\,c) = ((\mathbf{theIdentityElementOf}$ $G)\,(\mathbf{theBinaryOperationOf}G)\,c))$	DA BA 1.6, AS (0)

110	$(((\mathbf{theIdentityElementOf}G^{\cdot})\,(\mathbf{theBinaryOperationOf}G^{\cdot})\,c^{\cdot}) = c^{\cdot})$	DA BA 1.10, DA BA 1.5 (0, 1)
111	$(b^{\cdot} = c^{\cdot})$	AS (10, 11, 100, 101, 110)
1000	$\forall G((G\mathbf{isAGroup}) > \forall a\forall b\forall c(((b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G)) \wedge (c\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))) > (b = c)))$	>I, ∀I (0–111)

$\forall G((G\mathbf{isAGroup}) > \forall a\forall b\forall c(((b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G)) \wedge (c\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))) > (b = c)))$

1.4 The inverse element of an element in a group defining axiom for a function symbol

0	$\forall G((G\mathbf{isAGroup}) > \forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > \exists b(b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))))$	DA GT 1.1
1	$\forall G((G\mathbf{isAGroup}) > \forall a\forall b\forall c(((b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G)) \wedge (c\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))) > (b = c)))$	T GT 1.3
10	$\forall G((G\mathbf{isAGroup}) > \forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > \exists^{\times} b(b\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))))$	D∃×I (0, 1)

$\forall G((G\mathbf{isAGroup}) > \forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > ((\mathbf{theInverseElementOf}a\mathbf{In}G)\mathbf{isAnInverseElementOf}a\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G)\mathbf{Under}(\mathbf{theBinaryOperationOf}G)\mathbf{Of}(\mathbf{theUnderlyingSetOf}G))))$

1.5 The inverse of the inverse theorem

0	$((G^{\cdot}\mathbf{isAGroup}) \wedge (a^{\cdot} \ni (\mathbf{theUnderlyingSetOf}G^{\cdot})))$	
1	$((\mathbf{theInverseElementOf}a^{\cdot}\mathbf{In}G^{\cdot})\mathbf{isAnInverseElementOf}a^{\cdot}\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G^{\cdot})\mathbf{Under}(\mathbf{theBinaryOperationOf}G^{\cdot})\mathbf{Of}(\mathbf{theUnderlyingSetOf}G^{\cdot}))$	DA GT 1.4 (0)
10	$(a^{\cdot}\mathbf{isAnInverseElementOf}(\mathbf{theInverseElementOf}a^{\cdot}\mathbf{In}G^{\cdot})\mathbf{WithRespectTo}(\mathbf{theIdentityElementOf}G^{\cdot})\mathbf{Under}(\mathbf{theBinaryOperationOf}G^{\cdot})\mathbf{Of}(\mathbf{theUnderlyingSetOf}G^{\cdot}))$	DA BA 1.6, ∧E, ∧I (1)
11	$((\mathbf{theInverseElementOf}(\mathbf{theInverseElementOf}a^{\cdot}\mathbf{In}G^{\cdot})\mathbf{In}G^{\cdot}) = a^{\cdot})$	DA GT 1.4, T GT 1.3 (0, 1, 10)
100	$\forall G((G\mathbf{isAGroup}) > \forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > ((\mathbf{theInverseElementOf}(\mathbf{theInverseElementOf}a\mathbf{In}G)\mathbf{In}G) = a)))$	>I, ∀I (0–11)

$\forall G((G\mathbf{isAGroup}) > \forall a((a \ni (\mathbf{theUnderlyingSetOf}G)) > ((\mathbf{theInverseElementOf}(\mathbf{theInverseElementOf}a\mathbf{In}G)\mathbf{In}G) = a)))$

3 Topology

3.1 Topological Spaces

1.1 A set being a topology on a set defining axiom for a relation symbol

$$\forall X \forall \mathcal{T}((\mathcal{T}\ \mathbf{isATopologyOn}\ X) \diamond ((\mathcal{T} \supset \mathcal{P}(X)) \wedge ((\bigcirc \ni \mathcal{T}) \wedge ((X \ni \mathcal{T}) \wedge (\forall S((S \supset \mathcal{T}) > (\bigcup S \ni \mathcal{T})) \wedge \forall U \forall V(((U \ni \mathcal{T}) \wedge (V \ni \mathcal{T})) > ((U \cap V) \ni \mathcal{T})))))))$$

1.2 A set being a topological space defining axiom for a relation symbol

$$\forall P((P\ \mathbf{isATopologicalSpace}) \diamond ((P\ \mathbf{isATuple}) \wedge (((\mathbf{theLengthOf}\ P) = 2) \wedge ((\mathbf{theValueOf}\ P\ \mathbf{At} 1)\ \mathbf{isATopologyOn}\ (\mathbf{theValueOf}\ P\ \mathbf{At} 0)))))$$

1.3 The underlying set of a topological space defining axiom for a function symbol

0	$((\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0) = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0))$	=I
1	$\exists X(X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0))$	∃I (0)
10	$\forall X \forall Y(((X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0)) \wedge (Y = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 0))$	D∃×I (1, 10)

$$\forall P((P\ \mathbf{isATopologicalSpace}) > ((\mathbf{theUnderlyingSetOfTopologicalSpace}\ P) = (\mathbf{theValueOf}\ P\ \mathbf{At} 0)))$$

1.4 The topology of a topological space defining axiom for a function symbol

0	$((\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1) = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1))$	=I
1	$\exists X(X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1))$	∃I (0)
10	$\forall X \forall Y(((X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1)) \wedge (Y = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = (\mathbf{theValueOf}\ P^{\cdot}\ \mathbf{At} 1))$	D∃×I (1, 10)

$$\forall P((P\ \mathbf{isATopologicalSpace}) > ((\mathbf{theTopologyOf}\ P) = (\mathbf{theValueOf}\ P\ \mathbf{At} 1)))$$

1.5 A set being open in a topological space defining axiom for a relation symbol

$\forall P((P\textbf{isATopologicalSpace}) > \forall U((U\textbf{isAnOpenSetOf}P) \diamond (U \ni (\textbf{theTopologyOf}P))))$

1.6 A set being closed in a topological space defining axiom for a relation symbol

$\forall P((P\textbf{isATopologicalSpace}) > \forall C((C\textbf{isAClosedSetOf}P) \diamond ((C \supset (\textbf{theUnderlyingSetOfTopologicalSpace}P)) \wedge (((\textbf{theUnderlyingSetOfTopologicalSpace}P) \sim C) \ni (\textbf{theTopologyOf}P)))))$

1.7 The discrete topology on a set defining axiom for a function symbol

0	$(\mathcal{P}(X^{\cdot}) = \mathcal{P}(X^{\cdot}))$	=I
1	$\exists X(X = \mathcal{P}(X^{\cdot}))$	$\exists$I (0)
10	$\forall X \forall Y(((X = \mathcal{P}(X^{\cdot})) \wedge (Y = \mathcal{P}(X^{\cdot}))) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \mathcal{P}(X^{\cdot}))$	D$\exists^{\times}$I (1, 10)

$\forall X((\textbf{theDiscreteTopologyOn}X) = \mathcal{P}(X))$

1.8 The discrete topology is a topology theorem

0	$((\mathcal{P}(X^{\cdot}) \supset \mathcal{P}(X^{\cdot})) \wedge ((\bigcirc \ni \mathcal{P}(X^{\cdot})) \wedge (X^{\cdot} \ni \mathcal{P}(X^{\cdot}))))$	T 1.9, DA 1.13, DA 1.8, DA 1.25
1	$\forall S((S \supset \mathcal{P}(X^{\cdot})) > (\bigcup S \ni \mathcal{P}(X^{\cdot})))$	DA 1.20, DA 1.8, DA 1.25
10	$\forall U \forall V(((U \ni \mathcal{P}(X^{\cdot})) \wedge (V \ni \mathcal{P}(X^{\cdot}))) > ((U \cap V) \ni \mathcal{P}(X^{\cdot})))$	DA 1.22, DA 1.8, DA 1.25
11	$\forall X((\textbf{theDiscreteTopologyOn}X)\textbf{isATopologyOn}X)$	DA TOP 1.1, DA TOP 1.7, $\wedge$I, $\forall$I (0, 1, 10)

$\forall X((\textbf{theDiscreteTopologyOn}X)\textbf{isATopologyOn}X)$

1.9 The indiscrete topology on a set defining axiom for a function symbol

0	$(\{\bigcirc, X^{\cdot}\} = \{\bigcirc, X^{\cdot}\})$	=I
1	$\exists X(X = \{\bigcirc, X^{\cdot}\})$	∃I (0)
10	$\forall X \forall Y(((X = \{\bigcirc, X^{\cdot}\}) \wedge (Y = \{\bigcirc, X^{\cdot}\})) > (X = Y))$	AS, =I
11	$\exists^{\times} X(X = \{\bigcirc, X^{\cdot}\})$	D∃$^{\times}$I (1, 10)

$\forall X((\mathbf{theIndiscreteTopologyOn}X) = \{\bigcirc, X\})$

1.10 The indiscrete topology is a topology theorem

0	$((\{\bigcirc, X^{\cdot}\} \supset \mathcal{P}(X^{\cdot})) \wedge ((\bigcirc \ni \{\bigcirc, X^{\cdot}\}) \wedge (X^{\cdot} \ni \{\bigcirc, X^{\cdot}\})))$	DA 1.17, DA 1.25, T 1.9, DA 1.13
1	$\forall S((S \supset \{\bigcirc, X^{\cdot}\}) > (((X^{\cdot} \ni S) > (\bigcup S = X^{\cdot})) \wedge ((\backslash(X^{\cdot} \ni S) > (\bigcup S = \bigcirc)))))$	DA 1.17, DA 1.20, T 1.6
10	$\forall S((S \supset \{\bigcirc, X^{\cdot}\}) > (\bigcup S \ni \{\bigcirc, X^{\cdot}\}))$	DA 1.17, D\E, ∨E (1)
11	$\forall U \forall V(((U \ni \{\bigcirc, X^{\cdot}\}) \wedge (V \ni \{\bigcirc, X^{\cdot}\})) > ((U \cap V) \ni \{\bigcirc, X^{\cdot}\}))$	DA 1.17, DA 1.22, T 1.6, ∨E
100	$\forall X((\mathbf{theIndiscreteTopologyOn}X)\mathbf{isATopologyOn}X)$	DA TOP 1.1, DA TOP 1.9, ∧I, ∀I (0, 10, 11)

$\forall X((\mathbf{theIndiscreteTopologyOn}X)\mathbf{isATopologyOn}X)$